

High-Dimensional Instrumental Variable Quantile Regression under Weak Instruments: A Monte Carlo Study

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Supervisor:
JProf. Claudia Noack Ph.D.

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Olimjon Umurzokov

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Abstract

This thesis studies the finite-sample consequences of weakening instrument relevance in high-dimensional instrumental variable quantile regression (IVQR). Building on a preserved DML-IVQR Monte Carlo design, instrument strength is systematically varied through $\kappa \in \{1, 0.50, 0.25, 0.10\}$ while holding the marginal treatment distribution, structural outcome equation, and control dimension fixed. Three estimators—Oracle-GMM (an infeasible benchmark with known relevant controls), Full-GMM (unregularized estimation with all controls), and DML-IVQR (regularized nuisance estimation with pivotal penalty calibration)—are compared across five quantiles, $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$, and sample sizes $n \in \{500, 1000\}$.

The results show that as κ falls, point-estimation errors (MAE and RMSE) rise sharply, accepted sets expand, and power against false treatment-effect values declines for all estimators, including the oracle benchmark. Coverage is also not always well calibrated in some finite-sample cells, even under strong instrument relevance. DML often improves on Full-GMM but does not uniformly match Oracle-GMM across all design cells. The main conclusion is that Neyman orthogonalization and regularized nuisance estimation can reduce high-dimensional estimation costs, but they cannot replace identifying information lost when instruments become weakly relevant.

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1 Introduction

Economic effects are rarely well summarized by a single average. The impact of a treatment may differ substantially across the outcome distribution, while the treatment itself may be endogenous and the empirical specification may require a large set of controls. Instrumental variable quantile regression (IVQR) provides a framework for studying such heterogeneous structural effects under endogeneity, and double/debiased machine learning (DML) extends this framework to settings with high-dimensional nuisance parameters. However, even with high-dimensional controls and orthogonalization, weak instruments can still remain a serious problem. When identification deteriorates, point estimates can become unstable and confidence regions can lose informativeness even when inference remains formally valid. In this thesis we study that problem directly by examining how progressively weaker instruments affect the finite-sample performance and informativeness of weak-identification-robust DML-IVQR inference, and whether these effects differ across the outcome distribution.

Quantile treatment effects describe how a structural effect varies across quantiles of the potential-outcome distribution. When treatment is endogenous, conventional quantile regression does not generally recover these effects because treatment choices may be related to unobserved determinants of the outcome. IVQR solves this problem by combining excluded instruments with restrictions on the structural ranks of potential outcomes (Chernozhukov and Hansen 2005).

Two additional complications arise in modern applications. First, empirical specifications may contain many controls, transformations, and interactions. Their coefficients are nuisance parameters rather than the primary object of interest, but estimating a large nuisance model without structure can be unstable and may contaminate inference on the treatment effect. Regularization provides one way to estimate such high-dimensional components, while Neyman-orthogonal scores reduce the first-order sensitivity of the target moment to sufficiently small nuisance-estimation errors (Chernozhukov et al. 2018). Chen et al. (2021) combine these ideas with IVQR and develop a DML-IVQR procedure that handles high-dimensional controls while conducting candidate-specific inference on the structural quantile effect.

Second, regularization does not solve weak identification. Even a valid excluded instrument may contain little information about the endogenous treatment. When instrument-induced treatment variation is small, different candidate treatment-effect values can generate similar population moments, making them difficult to distinguish statistically. This problem is particularly important in IVQR because the relevant identifying restrictions are quantile specific and nonlinear. Standard first-stage measures indicate how strongly the instrument is

related to treatment, but they do not fully capture how much information the instrument provides for identifying a structural quantile effect. The broader weak-instrument literature shows why conventional estimate-centered approximations can become unreliable when identifying information is weak, and why candidate-specific procedures can remain useful even when point identification is imprecise (Staiger and Stock 1997; Stock and Wright 2000; Andrews et al. 2019). In IVQR, this distinction is reflected in weak-identification-robust test inversion: a procedure may remain asymptotically valid while producing broad sets of accepted candidate values and little power against economically relevant false alternatives.

High-dimensional nuisance estimation and weak instrument relevance are conceptually different problems. Neyman orthogonality reduces sensitivity to nuisance-estimation errors; it does not create identifying variation. Weak-identification-robust inference avoids relying on strong local identification, but cannot guarantee precise or informative confidence regions. Performance may therefore break down as instruments weaken even when nuisance estimation is handled well.

This distinction motivates the empirical question studied in this thesis. Existing Monte Carlo evidence for DML-IVQR evaluates regularized nuisance estimation in a high-dimensional design with a fixed level of instrument relevance. It therefore does not systematically show how the same estimators and inference procedures behave when excluded-instrument relevance is progressively weakened while the rest of the structural environment is held fixed. The central research question is: How does weakening instrument strength affect the finite-sample performance and informativeness of weak-identification-robust inference in DML-IVQR, and how do these effects vary across quantiles?

The thesis answers this question through a controlled extension of the Monte Carlo design in Chen et al. (2021). A single parameter, $\kappa \in \{1.00, 0.50, 0.25, 0.10\}$, controls the loading of the excluded-instrument components in the latent treatment equation. Smaller values reduce excluded-instrument relevance. An independent component is added to the treatment index to preserve its marginal variance as κ varies. The marginal treatment distribution, endogeneity, structural outcome equation, true quantile treatment effects, control dimension, and observed instrument construction are also preserved. This keeps the four relevance settings directly comparable while changing only how strongly the excluded instruments contribute to treatment. The parameter κ is a controlled relevance parameter for this Monte Carlo design, not a universal scalar measure of IVQR identification strength.

Three estimators are compared on identical simulated samples. Oracle-GMM serves as an infeasible benchmark because it knows the true set of relevant controls. Full-GMM uses the full high-dimensional control vector without regularization. DML-IVQR uses regular-

ized nuisance estimation together with the preserved Belloni–Chernozhukov penalty implementation. The comparison separates the common effect of declining instrument information from estimator-specific consequences of nuisance estimation, although differences between Full-GMM and DML-IVQR cannot be attributed to regularization alone because their residualization implementations also differ. Performance is evaluated across five quantiles, $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$, and two sample sizes, $n \in \{500, 1000\}$.

The evaluation separates point-estimation accuracy, coverage, power, and accepted-set informativeness. Bias, mean absolute error (MAE), and root mean squared error (RMSE) summarize accuracy. Coverage measures retention of the true structural quantile effect by the candidate-specific test, while power measures rejection of false treatment-effect values. Inferential informativeness is summarized by the grid-based accepted-set measure within the pre-specified primary computational domain $A_0 = [-1, 3]$; the choice of this domain and the corresponding domain-expansion diagnostic are explained in Appendix A.1. These distinctions matter: high coverage may accompany acceptance of many false candidates, while narrow accepted sets may reject the true value too frequently.

The Monte Carlo evidence shows a break down in informativeness as instrument relevance declines. Point-estimation error rises gradually for Oracle-GMM, Full-GMM, and DML-IVQR; accepted sets expand; and power against false candidate values falls. The deterioration of the oracle benchmark is particularly informative because it shows that the loss is not generated only by high-dimensional nuisance estimation. At the same time, the magnitude of the deterioration varies across quantiles and estimators. Coverage is also not uniformly close to its nominal level in finite samples, including in some strong-relevance cells. These results describe the implemented tests using a chi-square reference cutoff; the experiment does not establish the applicability of the asymptotic regularity conditions or identify the cause of undercoverage. DML-IVQR often performs better than the unregularized high-dimensional benchmark, but it does not dominate in every design cell and cannot compensate for the loss of identifying information when instruments become only weakly relevant. The main contribution of the thesis is therefore to document, within a controlled high-dimensional IVQR experiment, how nuisance robustness, instrument relevance, inferential informativeness, and finite-sample calibration interact as the information supplied by the excluded instruments is systematically reduced.

2 Literature Review

Quantile regression, introduced by Koenker and Bassett (1978), describes how covariates are related to different parts of a conditional outcome distribution rather than only to its mean. This is useful when treatment effects differ across the outcome distribution. Standard quantile regression, however, does not generally recover structural treatment effects when treatment is endogenous, because treatment may be related to unobserved factors that also affect the outcome. Chernozhukov and Hansen (2005) address this problem through instrumental variable quantile regression (IVQR). Their framework combines excluded instruments with restrictions on structural ranks to identify quantile-specific treatment effects in the presence of endogeneity.

A central feature of IVQR is the distinction between instrument validity and identification. Under the structural assumptions of Chernozhukov and Hansen (2005), instrument exogeneity and rank restrictions imply conditional moment conditions that can be used for estimation. Rank invariance assumes that an individual's structural rank remains the same across treatment states, while the weaker rank-similarity condition allows the rank to change as long as its conditional distribution is preserved. These assumptions connect observed outcomes to structural quantiles. They do not guarantee that the structural effect is precisely identified. The instrument must also provide enough treatment variation to distinguish the true structural effect from competing candidate values. Because IVQR moment conditions are quantile specific and nonlinear, conventional first-stage measures provide useful information about instrument relevance but do not fully describe identification at a particular quantile (Chernozhukov and Hansen 2005; Chernozhukov et al. 2017).

Chernozhukov and Hansen (2006) develop practical estimation and inference methods for the IVQR model under regular identification. Their inverse quantile regression approach considers candidate values of the endogenous-treatment coefficient and uses conventional quantile regression to estimate the remaining parameters. This reduces the difficult optimization problem to the lower-dimensional treatment parameter and provides a practical route for IVQR estimation. Chernozhukov and Hansen (2008) extend this approach to inference that is robust to weak identification. Instead of relying on a normal approximation centered on a precisely estimated treatment effect, their procedure tests each candidate treatment-effect value directly and constructs a confidence region by retaining the values that are not rejected. The resulting region can be wide, asymmetric, or disconnected when the instruments contain little identifying information. This is an important distinction for our thesis: weak-identification-robust inference may remain valid while providing little information about the value of the structural

effect. Chernozhukov et al. (2009) develop a different finite-sample conditional inference procedure for quantile regression models.

The IVQR weak-identification problem is related to the broader literature on weak instrumental variables. Anderson and Rubin (1949) provide an early approach based on testing structural parameter values without requiring a precisely estimated first stage. Staiger and Stock (1997) show why conventional instrumental-variable approximations can perform poorly when instruments are weak, while Stock and Wright (2000) develop weak-identification-robust methods for generalized method-of-moments (GMM) models. Kleibergen (2002) and Moreira (2003) provide further robust testing procedures, and Andrews et al. (2019) review the modern weak-instrument literature and its implications for empirical research. These contributions establish the importance of inference procedures that remain reliable when identification is weak. However, the usual linear-IV diagnostics do not provide a direct measure of quantile-specific IVQR identification. For this reason, we report the first-stage statistics to describe the ordering of instrument relevance across the Monte Carlo designs rather than formal IVQR weak-instrument thresholds.

A separate literature addresses inference with many controls or other high-dimensional nuisance parameters. Belloni and Chernozhukov (2011) study ℓ_1 -penalized quantile regression under approximate sparsity and develop a pivotal method for choosing the penalty level. This type of regularization makes estimation feasible when the number of possible controls is large, although regularization can itself introduce estimation error. Belloni et al. (2019) show how orthogonal score functions can support valid inference on a parameter of interest in high-dimensional approximately sparse quantile regression models. More generally, Chernozhukov et al. (2018) develop the double/debiased machine learning (DML) framework. Its central idea is to construct Neyman-orthogonal scores whose population moments are less sensitive, to first order, to small errors in estimated nuisance parameters. Additionally, Cross-fitting can reduce overfitting dependence by estimating nuisance functions and evaluating the score on different parts of the sample. Orthogonality and cross-fitting should therefore be distinguished: orthogonality is a property of the score, while cross-fitting is a way of organizing estimation across observations.

These high-dimensional methods address nuisance estimation rather than weak identification. In particular, regularization and Neyman orthogonality cannot make an instrument more relevant or create information that is absent from the data. Chen et al. (2021) combine IVQR with high-dimensional nuisance estimation by developing a DML-IVQR procedure based on a Neyman-orthogonal score. For each candidate treatment-effect value, nuisance parameters are estimated with regularization and the resulting score is used for GMM estima-

tion and candidate-specific inference. Their high-dimensional Monte Carlo design compares Oracle-GMM, which knows the relevant controls, Full-GMM, which uses all controls without regularization, and DML-IVQR, which uses regularized nuisance estimation. They also construct weak-identification-robust confidence regions for these estimators. Their results show that DML-IVQR can substantially improve on an unregularized high-dimensional specification and can approach the oracle benchmark in their baseline designs.

The implementation of Chen et al. (2021) is especially important for our thesis because it provides the baseline Monte Carlo environment that is preserved and extended. Their main high-dimensional simulation considers a fixed level of instrument relevance. It therefore does not systematically examine how the same estimators and candidate-specific inference procedures behave as instrument relevance becomes progressively weaker. This distinction matters because successful high-dimensional nuisance estimation does not imply that the structural parameter remains well identified. A DML procedure may estimate nuisance components accurately while the instruments still provide too little information to distinguish competing structural effects.

Other work illustrates continuing developments in IVQR estimation and high-dimensional quantile methods. Wüthrich (2019) develops an approach for estimating quantile treatment effects with endogenous binary treatment and instruments, providing an alternative to directly optimizing the standard IVQR objective. More recently, Jin and Sun (2025) develop a Neyman-orthogonal moment for a nonparametric IVQR model with high-dimensional covariates and focus on unconditional quantile treatment effects. They recommend cross-fitting and automatic DML methods for their setting. These approaches differ from the linear structural IVQR model and preserved Monte Carlo implementation studied in our thesis.

The literature therefore leaves a specific finite-sample question that is central to our thesis. Existing works provide structural IVQR model, weak-identification-robust inference, high-dimensional regularization, and orthogonal-score methods, but there is a limited evidence on how weak instrument relevance affects these components jointly in the preserved high-dimensional DML-IVQR design of Chen et al. (2021). In our thesis we address that question by systematically reducing instrument relevance while keeping the remaining features of the Monte Carlo environment as stable as possible. It evaluates how point-estimation accuracy, coverage, power, and accepted-set informativeness change across instrument-relevance settings, quantiles, and estimators. The contribution is therefore a controlled finite-sample extension of the existing DML-IVQR design of Chen et al. (2021).

3 Methodology

3.1 Structural IVQR Model and Identification

Following the IVQR framework of Chernozhukov and Hansen (2005), let Y_i denote the outcome, D_i the endogenous treatment or vector of endogenous variables, X_i a vector of observed controls, and Z_i a vector of instruments. The structural quantile function is denoted by $q(u, d, x)$, where $u \in (0, 1)$ represents the structural rank. For a fixed treatment state d and covariates x , $q(\tau, d, x)$ gives the structural τ -quantile of the potential outcome under treatment state d .

Under the IVQR assumptions on instrument exogeneity, treatment selection, and rank similarity, the observed data can be represented as

$$Y_i = q(U_i, D_i, X_i), \quad U_i \mid X_i, Z_i \sim \text{Uniform}(0, 1), \quad (1)$$

where $U_i := U_{D_i i}$ is the realized structural rank. Rank similarity allows treatment-state-specific ranks to differ while requiring their conditional distributions to be the same across treatment states. Rank invariance is the stronger special case in which an individual's rank is unchanged across treatment states. Strict monotonicity of $q(u, d, x)$ in u links the structural rank to the corresponding structural quantile.

Treatment endogeneity arises because the realized treatment D_i may depend on unobserved factors that are related to the structural rank. Consequently, the observed conditional quantile of Y_i given D_i and X_i does not generally equal the structural quantile obtained under a fixed treatment state. IVQR uses instruments to recover the latter object. For two treatment states d_1 and d_0 , the structural quantile treatment effect at quantile τ and covariates x is

$$q(\tau, d_1, x) - q(\tau, d_0, x).$$

For a fixed quantile index $\tau \in (0, 1)$, the DML-IVQR framework considered in Chen et al. (2021) uses the linear-in-parameters structural quantile specification

$$q(\tau, D_i, X_i) = c_0(\tau) + D_i' \alpha_0(\tau) + X_i' \beta_0(\tau). \quad (2)$$

The parameter of interest is $\alpha_0(\tau)$, the structural treatment coefficient at quantile τ . The intercept $c_0(\tau)$ and control coefficients $\beta_0(\tau)$ are nuisance parameters. Their specific values

in the Monte Carlo data-generating process are introduced later when the simulation design is defined.

The key IVQR restriction follows from the structural-rank representation. Because $q(u, d, x)$ is strictly increasing in u ,

$$Y_i \leq q(\tau, D_i, X_i) \iff U_i \leq \tau.$$

Since the realized structural rank is conditionally uniform given X_i and Z_i ,

$$\Pr(U_i \leq \tau \mid X_i, Z_i) = \tau.$$

Substituting the linear structural quantile specification gives

$$\Pr[Y_i \leq c_0(\tau) + D_i' \alpha_0(\tau) + X_i' \beta_0(\tau) \mid X_i, Z_i] = \tau. \quad (3)$$

Let $\mathbf{1}\{A\}$ denote the indicator function, equal to one when statement A is true and zero otherwise. The same restriction can then be written as the conditional moment condition

$$\mathbb{E}[\tau - \mathbf{1}\{Y_i \leq c_0(\tau) + D_i' \alpha_0(\tau) + X_i' \beta_0(\tau)\} \mid X_i, Z_i] = 0. \quad (4)$$

Equation (4) is the central population restriction used by IVQR. Instrument exogeneity justifies the conditional moment restriction, while identification requires the instruments to provide enough variation to distinguish the true structural coefficient from competing candidate values. These are separate requirements: an instrument may satisfy the validity conditions but still provide too little information for precise identification of $\alpha_0(\tau)$. Section 3.2 builds on this restriction and constructs the nuisance-robust score used by DML-IVQR.

3.2 Orthogonal Score and Nuisance Estimation

Fix a quantile τ and consider a candidate value a for the structural treatment coefficient $\alpha_0(\tau)$. IVQR evaluates such candidate values one at a time. For a given a , subtracting $D_i' a$ from the outcome leaves an adjusted outcome whose conditional τ -quantile must be explained by an intercept and the controls. The first nuisance-estimation problem therefore profiles these quantities conditional on the candidate a .

Define the augmented control vector

$$\tilde{X}_i = (1, X_i')',$$

where the first component is the constant and the remaining components are the p nonconstant controls. Quantile regression measures fit with the check-loss function

$$\rho_\tau(u) = u[\tau - \mathbf{1}\{u < 0\}]. \quad (5)$$

Here u is a generic scalar residual argument of the check-loss function. The check loss is asymmetric: observations above and below the candidate quantile receive weights determined by τ . Minimizing this loss therefore fits the conditional τ -quantile rather than the conditional mean.

Following Chen et al. (2021), for each candidate a , let $c(a)$ denote the profiled intercept and let $\beta(a)$ denote the profiled vector of control slopes. They are defined jointly by

$$\begin{pmatrix} c(a) \\ \beta(a) \end{pmatrix} = \arg \min_{c,b} \mathbb{E}[\rho_\tau(Y_i - D_i' a - c - X_i' b)]. \quad (6)$$

Here $c \in \mathbb{R}$ is the optimization variable for the intercept and $b \in \mathbb{R}^p$ is the optimization vector for the control slopes. Their minimizers are $c(a)$ and $\beta(a)$. These are candidate-specific nuisance parameters: changing a changes the adjusted outcome $Y_i - D_i' a$, and therefore generally changes the fitted intercept and control coefficients. At the true candidate $a = \alpha_0(\tau)$, maintained uniqueness gives $c(a) = c_0(\tau)$ and $\beta(a) = \beta_0(\tau)$.

$$\mathbb{E}[\{\tau - \mathbf{1}[Y_i \leq D_i' a + c(a) + X_i' \beta(a)]\} \tilde{X}_i] = 0. \quad (7)$$

The moment above is the first-order quantile-regression condition for the profiled intercept and controls. It says that, at the fitted candidate-specific quantile, the quantile residual is orthogonal to the augmented controls.

With the candidate treatment effect and profiled nuisance coefficients fixed, define the candidate residual

$$\epsilon_i(a) = Y_i - D_i' a - c(a) - X_i' \beta(a). \quad (8)$$

The residual equals zero at the candidate fitted quantile. The local behavior of a quantile moment depends on how much probability mass lies near this zero boundary. This is why the residual density at zero enters the orthogonalization. Let $f_{\epsilon(a)}(0 \mid X_i, Z_i)$ be its conditional density at zero.

Suppose $Z_i \in \mathbb{R}^{d_z}$ and $X_i \in \mathbb{R}^p$, so $\tilde{X}_i \in \mathbb{R}^{p+1}$. The next step removes from the instruments the component that is locally associated with the controls. Because quantile moments change according to the residual density at zero, this residualization is density weighted rather

than an ordinary unweighted linear projection. Define

$$\begin{aligned} M(a) &= E[Z_i \tilde{X}_i' f_{\epsilon(a)}(0 \mid X_i, Z_i)], \quad M(a) \in \mathbb{R}^{d_z \times (p+1)}, \\ J(a) &= E[\tilde{X}_i \tilde{X}_i' f_{\epsilon(a)}(0 \mid X_i, Z_i)], \quad J(a) \in \mathbb{R}^{(p+1) \times (p+1)}. \end{aligned} \quad (9)$$

Assuming $J(a)$ is nonsingular, set

$$\delta(a) = M(a)J(a)^{-1}, \quad \delta(a) \in \mathbb{R}^{d_z \times (p+1)}, \quad (10)$$

Thus $J(a)$ describes density-weighted variation in the augmented controls, while $M(a)$ describes their density-weighted relationship with the instruments. The matrix $\delta(a) = M(a)J(a)^{-1}$ contains the coefficients of the corresponding population residualization. Partition this matrix as $\delta(a) = [d_0(a), \Gamma(a)]$, where $d_0(a) \in \mathbb{R}^{d_z}$ contains the residualization intercepts and $\Gamma(a) \in \mathbb{R}^{d_z \times p}$ contains the slopes on the nonconstant controls. The corresponding population residualized instrument is

$$\Psi_i(a) = Z_i - \delta(a)\tilde{X}_i = Z_i - d_0(a) - \Gamma(a)X_i. \quad (11)$$

The vector $\Psi_i(a)$ is therefore the instrument after removing the density-weighted component associated with the intercept and controls. It is this residualized instrument, rather than the raw Z_i , that enters the theoretical orthogonal score. By construction, $E[(Z_i - \delta(a)\tilde{X}_i)\tilde{X}_i' f_{\epsilon(a)}(0 \mid X_i, Z_i)] = 0$; this is not an ordinary unweighted residual or a conventional first-stage coefficient.

For the simulation DGP in Section 3.5, this idealized population construction requires a qualification. At the true structural parameter, write $e_\tau = D(u - q_\tau)$, where $q_\tau = \Phi^{-1}(\tau)$ and $D = \Phi(T)$ with $T = d^*(\kappa)$. Conditional on $u = q_\tau$, X , and the observed instruments Z , T is normal with variance $2.91 - \kappa^2 > 1$ for every investigated κ . The population residual density at zero contains a term proportional to $E[1/\Phi(T) \mid u = q_\tau, X, Z]$, which is not finite under these conditional normal variances. The usual finite-density-matrix regularity argument for $J(a)$ and $M(a)$ is therefore not established for this DGP. This thesis does not claim that the preserved executable implementation satisfies the complete regularity conditions underlying the idealized asymptotic orthogonality argument; the study is a finite-sample evaluation of that executable procedure. This limitation neither proves that its statistic is invalid nor establishes the cause of observed undercoverage.

The candidate-specific nuisance vector collects the quantities required to construct the score.

$$\eta(a) = \begin{pmatrix} c(a) \\ \beta(a) \\ \text{vec}\{\delta(a)\} \end{pmatrix}, \quad \eta_0 = \eta(\alpha_0(\tau)) = \begin{pmatrix} c_0(\tau) \\ \beta_0(\tau) \\ \text{vec}\{\delta(\alpha_0(\tau))\} \end{pmatrix}. \quad (12)$$

Vectorization only stacks the residualization matrix. The target remains $\alpha_0(\tau)$; $c(a)$ and $\beta(a)$ profile the outcome nuisance, while $\delta(a)$ residualizes the instruments.

Following Chen et al. (2021), the orthogonal score combines two objects: the candidate quantile residual and the density-weighted residualized instrument.

$$\begin{aligned} g_\tau(W_i; a, \eta) &= [\tau - \mathbf{1}\{Y_i \leq D_i' a + c + X_i' \beta\}] (Z_i - \delta \tilde{X}_i), \\ g_\tau(W_i; a, \eta(a)) &= [\tau - \mathbf{1}\{Y_i \leq D_i' a + c(a) + X_i' \beta(a)\}] \Psi_i(a), \end{aligned} \quad (13)$$

where $W_i = (Y_i, D_i, X_i, Z_i)$. The first factor asks whether the observed outcome lies below the candidate structural quantile. The second factor supplies instrument variation after removing the control-related component. At the true treatment coefficient and true nuisance values, their population product has expectation zero.

At $a = \alpha_0(\tau)$ and $\eta = \eta_0$, the score satisfies

$$\mathbb{E}[g_\tau(W_i; \alpha_0(\tau), \eta_0)] = 0. \quad (14)$$

Ordinary plug-in estimation of high-dimensional nuisances can contaminate a target moment. DML therefore uses a score with the stronger local property that its population expectation is insensitive, to first order, to small perturbations of the nuisance vector at the truth. Formally,

$$\partial_\eta \mathbb{E}[g_\tau(W_i; \alpha_0(\tau), \eta)]|_{\eta=\eta_0} = 0. \quad (15)$$

Here ∂_η denotes differentiation with respect to the nuisance vector. Orthogonality does not mean that nuisance estimation is error free. It means that sufficiently small nuisance-estimation errors do not enter the target moment at first order.

The cancellation can be seen directly. First consider perturbations of the profiled intercept and control slopes. To show the cancellation for the intercept and slopes, write $\epsilon_{0i} = \epsilon_i(\alpha_0(\tau))$ and define

$$\begin{aligned} M_0 &= \mathbb{E}[Z_i \tilde{X}_i' f_{\epsilon_0}(0 | X_i, Z_i)], \\ J_0 &= \mathbb{E}[\tilde{X}_i \tilde{X}_i' f_{\epsilon_0}(0 | X_i, Z_i)], \\ \delta_0 &= M_0 J_0^{-1}. \end{aligned} \quad (16)$$

At the truth, differentiation with respect to the fitted intercept and control slopes then gives

$$\begin{aligned}\partial_{(c,\beta')} E[g_\tau(W_i; \alpha_0(\tau), \eta)]|_{\eta=\eta_0} &= -E[(Z_i - \delta_0 \tilde{X}_i) \tilde{X}_i' f_{\epsilon_0}(0 | X_i, Z_i)] \\ &= -[M_0 - \delta_0 J_0] = 0.\end{aligned}\tag{17}$$

The density appears because observations near the boundary can switch sides. The choice $\delta_0 = M_0 J_0^{-1}$ makes the residualized instrument density-weighted orthogonal to $\tilde{X}_i$, cancelling first-order changes in the intercept and slopes.

A similar argument applies to the instrument-residualization nuisance. For instrument component $j = 1, \dots, d_z$, let δ_j denote the corresponding row of δ . Then

$$\partial_{\delta_j} E[g_{\tau,j}(W_i; \alpha_0(\tau), \eta)]|_{\eta=\eta_0} = -E[\{\tau - \mathbf{1}[Y_i \leq D_i' \alpha_0(\tau) + c_0(\tau) + X_i' \beta_0(\tau)]\} \tilde{X}_i'] = 0.\tag{18}$$

An error in δ is multiplied by the quantile residual, whose moment with $\tilde{X}_i$ is zero by (7), so its first-order effect also vanishes. This construction permits sufficiently accurate regularized nuisances to support inference on the low-dimensional target (Chernozhukov et al. 2018; Chen et al. 2021).

Orthogonality protects the score against small nuisance-parameter errors, but it does not create identification or add information about the treatment effect. In particular, it does not make a weak instrument strong and does not guarantee that the data contain enough information about $\alpha_0(\tau)$. If the instruments contain little information about the endogenous treatment, the moment may also contain little information for distinguishing nearby values of a . IVQR moment validity comes from the structural assumptions and instrument exogeneity; nuisance robustness comes from orthogonalization. These are different requirements.

3.3 Estimators and Executable Implementation

Throughout the thesis, DML-IVQR refers to the implementation using the Belloni–Chernozhukov pivotal penalty described below.

Throughout this methodology section, N denotes the sample size. Section 3.5 uses n for the corresponding Monte Carlo design sample size. Section 3.2 defined population nuisances. When controls are numerous relative to N , DML-IVQR estimates their sample counterparts with ℓ_1 regularization under approximate sparsity: a relatively small group has substantial predictive content while other coefficients are zero or small.

Fix τ and a candidate value a . Nuisance estimation is repeated whenever a relevant candidate is evaluated. In the executable implementation used here, the response for the first nui-

sance step is $Y_i - D_i' a$. The formula adds an intercept, so let b_0 denote that intercept and let $b = (b_1, \dots, b_p)'$ contain only the slopes on X_i . Thus, $(\widehat{b}_0(a), \widehat{\beta}(a))$ estimates $(c(a), \beta(a))$. The implementation-specific notation for the later instrument regression is defined below and is kept distinct from the full population matrix $\delta(a) = [d_0(a), \Gamma(a)]$. The candidate-specific quantile-Lasso estimates are

$$\begin{pmatrix} \widehat{b}_0(a) \\ \widehat{\beta}(a) \end{pmatrix} = \arg \min_{b_0, b} \left\{ \sum_{i=1}^N \rho_\tau(Y_i - D_i' a - b_0 - X_i' b) + \frac{1}{2} \left[\lambda_0^{rq}(a) |b_0| + \sum_{k=1}^p \lambda_k^{rq}(a) |b_k| \right] \right\}. \quad (19)$$

The first term measures fit at quantile τ . The second places a separate penalty on each coefficient, including the intercept; there is no post-Lasso refit. The loss is an unscaled sum. The factor $1/2$ reflects the augmented-observation implementation of `rq.fit.lasso` in the preserved `quantreg 5.34` environment. Accordingly, $\lambda^{rq}(a)$ denotes the vector supplied to the routine rather than the coefficient multiplying the absolute-value penalty directly. The implementation uses the pivotal quantile-Lasso calibration developed by Belloni and Chernozhukov (2011), following the executable route preserved by Chen et al. (2021). Let $s_k = (N^{-1} \sum_i X_{ik}^2)^{1/2}$ and, for $r = 1, \dots, 1000$, draw independent $U_{ir} \sim \text{Uniform}(0, 1)$. The vector supplied to the routine is

$$\begin{aligned} T_r &= \max_{1 \leq k \leq p} \left| \frac{\sum_{i=1}^N X_{ik} [\tau - \mathbf{1}\{U_{ir} < \tau\}]}{s_k \sqrt{\tau(1-\tau)}} \right|, \\ \lambda^{rq}(a) &= 2 \widehat{Q}_{0.9}(\{T_r\}_{r=1}^{1000}) \sqrt{\tau(1-\tau)} (1, s_1, \dots, s_p)'. \end{aligned} \quad (20)$$

Here $k = 1, \dots, p$ indexes the control variables, $r = 1, \dots, 1000$ indexes the simulated pivotal draws, T_r is the simulated pivotal statistic in draw r , and $\widehat{Q}_{0.9}(\{T_r\}_{r=1}^{1000})$ denotes their empirical 90th percentile. The $p + 1$ entries of $\lambda^{rq}(a)$ correspond respectively to the intercept and the p control slopes. Artificial Uniform variables are useful because the indicator at the correct quantile has a known Bernoulli structure. The simulated statistics approximate how large random correlations between the controls and a quantile score can be without a true signal. The simulated 0.9 quantile sets the reference level for these random correlations, while the factor 2 scales the resulting penalty according to the implemented pivotal rule. A fresh set of Uniform draws is generated every time the penalty routine is called. Hence, candidate evaluations on the same observed sample can use different pivotal draws. This is penalty-construction randomness, not sample splitting. The frozen main simulation uses this rule rather than its older five-fold cross-validation alternative.

The fitted quantile regression produces the candidate residual and density weight

$$\begin{aligned}\widehat{\epsilon}_i(a) &= Y_i - D_i' a - \widehat{b}_0(a) - X_i' \widehat{\beta}(a), \\ \widehat{f}_i(a) &= \phi\left(\widehat{\epsilon}_i(a); \bar{\epsilon}(a), s_{\bar{\epsilon}(a)}^2\right),\end{aligned}\tag{21}$$

Here $\bar{\epsilon}(a)$ denotes the sample mean of the N candidate residuals, $s_{\bar{\epsilon}(a)}^2$ denotes their sample variance, and $\phi(e; \mu, \sigma)$ is the normal density with mean μ and standard deviation σ . Thus, the executable simulation forms the weights by evaluating a fitted normal density at the candidate residuals.¹ This fitted-normal quantity is an implementation-specific proxy for the theoretical conditional density $f_{\epsilon(a)}(0 \mid X_i, Z_i)$ from Section 3.2; it is not a direct conditional-density estimate at zero. Preserving this proxy is part of the replication-extension design; no consistency claim is made that it recovers the ideal population density object in this DGP. Zero remains the fitted quantile boundary, and the theoretical reason for density weighting is that observations near this boundary determine local changes in the quantile moment. These proxy weights are then used in the instrument-residualization step, where they play the role assigned to density weights in the theoretical construction.

For each instrument component $j = 1, \dots, d_z$, the implementation next fits a separate square-root-density-weighted Lasso. Let $\widehat{r}_{j0}(a)$ denote its retained ordinary intercept and let $\widehat{\Gamma}_{\text{imp},j}(a)$ contain only the slopes on the transformed controls. Its transformed regression is

$$\begin{pmatrix} \widehat{r}_{j0}(a) \\ \widehat{\Gamma}_{\text{imp},j}(a) \end{pmatrix} = \arg \min_{(r_{j0}, \gamma_j)} \left\{ \sum_{i=1}^N \left[\sqrt{\widehat{f}_i(a)} Z_{ji} - r_{j0} - \sqrt{\widehat{f}_i(a)} X_i' \gamma_j \right]^2 + \mathcal{P}_{j,a}(\gamma_j) \right\}.\tag{22}$$

Here $\mathcal{P}_{j,a}(\gamma_j) = \sum_{k=1}^p \lambda_{jk}^\Gamma(a) |\gamma_{jk}|$ is the coefficient-specific ℓ_1 penalty produced by the instrument-regression routine's default iterative heteroskedastic loadings. Here $\gamma_j \in \mathbb{R}^p$ is the slope vector for instrument component j , and $k = 1, \dots, p$ indexes its control coefficients. The intercept is included but not penalized, the same controls are used as in the first nuisance step, and no post-Lasso refit is performed. After fitting the transformed regression, the code retains both the intercept and the slopes. Define $\widehat{r}_0(a) = (\widehat{r}_{10}(a), \dots, \widehat{r}_{d_z 0}(a))'$ and stack the slope rows in $\widehat{\Gamma}_{\text{imp}}(a)$. The residualized instrument used in the score is

$$\begin{aligned}\widehat{\Psi}_{ji}(a) &= Z_{ji} - \widehat{r}_{j0}(a) - X_i' \widehat{\Gamma}_{\text{imp},j}(a), \\ \widehat{\Psi}_i(a) &= Z_i - \widehat{r}_0(a) - \widehat{\Gamma}_{\text{imp}}(a) X_i.\end{aligned}\tag{23}$$

1. The literal preserved call is `dnorm(e, mean(e), var(e))`. Its third argument is interpreted as a standard deviation, although the code supplies the residual sample variance.

Thus, the score residualizes the original, untransformed instrument. The calculation subtracts the retained intercept and the fitted contribution of the original controls. It does not divide the retained intercept by $\sqrt{\widehat{f}_i(a)}$ or use the transformed regression residual directly.

The intercept retained by the frozen transformed regression is implementation specific. It enters as an ordinary constant rather than $\sqrt{\widehat{f}_i(a)}$ times a constant regressor, so $\widehat{r}_0(a)$ is not the literal sample counterpart of $d_0(a)$ in the augmented population projection of Section 3.2. The exact derivative cancellation in Section 3.2 applies to the theoretical augmented score. The executable score follows its density-weighted slope residualization but treats the intercept differently. The results therefore characterize the preserved implementation, not a literal sample implementation of every component of the theoretical score.

The penalties can shrink weak coefficients to zero and stabilize both high-dimensional steps, but also create shrinkage error. The theoretical score in Section 3.2 limits its first-order effect (Chernozhukov et al. 2018). Nuisances and scores use the same sample; this preserved implementation has no sample splitting or cross-fitting.

Oracle-GMM, Full-GMM, and DML-IVQR target the same $\alpha_0(\tau)$ using the same outcome, treatment, instruments, and quantile. Each multiplies a candidate quantile score by residualized instruments, producing a moment with dimension d_z ; nuisance estimation differs.

Oracle-GMM is an infeasible benchmark because it is given the identity of the ten controls designated as relevant in the simulation DGP. Let $S_0 = \{1, \dots, 10\}$ denote this oracle control set and let $X_{S_0,i}$ collect these variables. At each candidate, the executable estimator uses an unpenalized quantile regression with an intercept:

$$\begin{pmatrix} \widehat{b}_{0,\text{orc}}(a) \\ \widehat{\beta}_{\text{orc}}(a) \end{pmatrix} = \arg \min_{b_0, b} \sum_{i=1}^N \rho_\tau(Y_i - D_i' a - b_0 - X_{S_0,i}' b). \quad (24)$$

Thus, Oracle-GMM first removes $D_i' a$ and then fits the remaining outcome at quantile τ using only this known oracle control set. No coefficient is penalized and there is no later variable-selection or refitting step. The oracle is not a practically available estimator and is not free from sampling error. It provides a benchmark for the cost of having to learn which controls matter.

Full-GMM does not receive this knowledge. It uses all available controls in the same unpenalized, candidate-specific quantile-regression problem:

$$\begin{pmatrix} \widehat{b}_{0,\text{full}}(a) \\ \widehat{\beta}_{\text{full}}(a) \end{pmatrix} = \arg \min_{b_0, b} \sum_{i=1}^N \rho_\tau(Y_i - D_i' a - b_0 - X_i' b). \quad (25)$$

Thus all 100 controls enter freely. Full-GMM does not require the unknown oracle set, but estimating many weak or irrelevant coefficients can be noisy.

Oracle-GMM and Full-GMM use the same unregularized instrument residualization with their respective control matrices. For $e \in \{\text{orc}, \text{full}\}$, let $X_{\text{orc},i} = X_{S_0,i}$ and $X_{\text{full},i} = X_i$, and let $\widehat{f}_{e,i}(a)$ be the fitted-normal proxy defined above. The code computes

$$\begin{aligned}\widehat{M}_e(a) &= \frac{1}{N} \sum_{i=1}^N Z_i X'_{e,i} \widehat{f}_{e,i}(a), & \widehat{J}_e(a) &= \frac{1}{N} \sum_{i=1}^N X_{e,i} X'_{e,i} \widehat{f}_{e,i}(a), \\ \widehat{\delta}_e(a) &= \widehat{M}_e(a) \widehat{J}_e(a)^{-1}, & \widehat{\Psi}_{e,i}(a) &= Z_i - \widehat{\delta}_e(a) X_{e,i}.\end{aligned}\tag{26}$$

This direct, unpenalized inversion includes no residualization intercept, although the quantile regression has one, and uses the original Z_i in the final residual. It therefore differs from the augmented population orthogonalization in Section 3.2. There is no ridge adjustment or generalized inverse: a singular $\widehat{J}_e(a)$ produces a recorded numerical failure. Each benchmark multiplies $\widehat{\Psi}_{e,i}(a)$ by its candidate quantile score.

DML-IVQR uses all controls and regularizes both nuisance steps. It estimates the control coefficients by quantile Lasso, forms the density proxy, and uses regularized weighted instrument regressions. DML-IVQR applies (20). Oracle and Full also use residualized IVQR moments, but estimate nuisances without ℓ_1 regularization. Unlike their matrix calculation, DML retains a residualization intercept. The comparison covers a known oracle set without regularization, all controls without regularization, and all controls with regularized nuisance estimation. It does not isolate every implementation difference but keeps the structural model, target, and instruments common.

3.4 GMM Criterion and Weak-Identification-Robust Inference

Let $e \in \{\text{orc}, \text{full}, \text{dml}\}$ index the estimator and $\widehat{g}_{e,i}(a) \in \mathbb{R}^{d_z}$ its observation-level score. Its sample average is

$$\widehat{g}_{e,N}(a) = \frac{1}{N} \sum_{i=1}^N \widehat{g}_{e,i}(a).\tag{27}$$

This vector measures the sample restriction at a . The code measures its variation with the uncentered second moment

$$\widehat{\Sigma}_e(a) = \frac{1}{N} \sum_{i=1}^N \widehat{g}_{e,i}(a) \widehat{g}_{e,i}(a)'. \tag{28}$$

This matrix standardizes differently scaled and correlated components. It uses denominator N , no finite-sample correction, and no sample-mean centering. At the truth, its probability limit equals the score covariance because the population moment is zero.

The common candidate-specific GMM criterion is

$$W_{e,N}(a) = N\widehat{g}_{e,N}(a)' \widehat{\Sigma}_e(a)^{-1} \widehat{g}_{e,N}(a). \quad (29)$$

This equals the code's calculation from the unnormalized score sum and sum of outer products. Smaller values indicate moments closer to zero after standardization. The direct inverse has no ridge or generalized inverse; failure or a nonfinite result is recorded. Suppressing e gives $W_N(a)$. Let $\mathcal{A}$ denote the set of candidate treatment-effect values over which the GMM profile is evaluated; its finite numerical specification in the Monte Carlo experiment is given in Section 3.7. The point estimator is

$$\widehat{\alpha}_e(\tau) = \arg \min_{a \in \mathcal{A}} W_{e,N}(a). \quad (30)$$

The minimum need not be zero. The implementation minimizes over the finite profile in Section 3.7, selects the first exact tie, and gives no point estimate to an incomplete profile. Point estimators are selected on the fixed numerical grid; except at quantiles whose true effect lies exactly on a grid point, reported point-estimation errors therefore also contain a small discretization component.

Point estimation retains the minimizer; inference tests each a separately. Under the regularity conditions for the corresponding theoretical candidate-specific IVQR score, the null statistic has the asymptotic reference distribution

$$W_{e,N}(a) \xrightarrow{d} \chi_{d_z}^2, \quad d_z = \dim(Z_i). \quad (31)$$

The degrees of freedom equal the number of moment components. The preserved Monte Carlo implementation applies this chi-square law as the asymptotic reference distribution for the candidate-specific statistics of all three estimators. For DML-IVQR, however, the implementation-specific residualization intercept described in Section 3.3 is not the literal sample counterpart of the augmented theoretical intercept. Together with the density and DGP qualifications in Sections 3.2–3.3, this means that the thesis does not establish the stated limit law for the preserved executable procedure. Following Chernozhukov and Hansen (2008), the test evaluates the hypothesized value directly rather than using a normal approximation for $\widehat{\alpha}_e(\tau)$. At level γ , reject when $W_{e,N}(a) > c_{1-\gamma, d_z}$. Here $\gamma \in (0, 1)$ is the test significance level,

and $c_{1-\gamma, d_z}$ is the $(1-\gamma)$ -quantile of a chi-square distribution with d_z degrees of freedom. This test-level γ is distinct from the instrument-regression slope vector γ_j used in Section 3.3.

Inverting these candidate-specific tests gives

$$C_{e,1-\gamma}(\tau) = \{a \in \mathcal{A} : W_{e,N}(a) \leq c_{1-\gamma, d_z}\}. \quad (32)$$

Test inversion retains every candidate not rejected by its own moment test. Because the profile can cross the cutoff repeatedly, the region may be connected, disconnected, wide, or empty. A finite numerical domain describes only acceptance within that domain and cannot establish global unboundedness.

The underlying Chernozhukov–Hansen test-inversion approach can retain asymptotic validity under weak, partial, or failed identification, subject to its regularity conditions, but need not be informative. Weak instruments can leave many candidates below the cutoff. Because the implementation uses an asymptotic chi-square critical value, finite-sample calibration remains empirical. This differs from the exact conditional simulation method of Chernozhukov et al. (2009).

3.5 Monte Carlo Design and Instrument-Relevance Extension

Sections 3.1–3.4 use N as generic sample-size notation, while Sections 3.5–3.7 and Chapter 4 use n for the Monte Carlo design sample size.

The Monte Carlo experiment builds on the high-dimensional simulation design of Chen et al. (2021), with $p = 100$ observed controls. For each observation i , the structural disturbances satisfy

$$\begin{pmatrix} u_i \\ \epsilon_i \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0.3 \\ 0.3 & 1 \end{pmatrix}\right). \quad (33)$$

The disturbance ϵ_i enters the treatment equation, whereas u_i enters the outcome through $D_i u_i$. Their positive correlation therefore generates endogeneity.

The remaining primitives are mutually independent and independent of the disturbance pair in Equation (33), and they satisfy

$$x_{ji} \stackrel{\text{ind}}{\sim} \mathcal{N}(0, 1), \quad j = 1, \dots, 100, \quad z_{1i}, z_{2i}, v_{1i}, v_{2i} \stackrel{\text{ind}}{\sim} \mathcal{N}(0, 1). \quad (34)$$

Observations are independent across i . The observed controls are

$$X_{ji} = \Phi(x_{ji}), \quad j = 1, \dots, 100, \quad (35)$$

The observed control vector is $X_i = (X_{1i}, \dots, X_{100i})'$. Here Φ is the standard-normal cumulative distribution function, so $X_{ji} \sim \mathcal{U}(0, 1)$ independently across controls.

The executable design constructs two instruments as²

$$Z_{1i} = z_{1i} + v_{1i} + X_{2i} + X_{3i} + X_{4i}, \quad (36)$$

$$Z_{2i} = z_{2i} + v_{2i} + X_{7i} + X_{8i} + X_{9i} + X_{10i}. \quad (37)$$

Thus $Z_i = (Z_{1i}, Z_{2i})'$, so the number of instrument components is $d_z = 2$. The z components enter the treatment index directly, the included controls induce dependence between instruments and controls, and the v components add independent noise. The executable baseline treatment is

$$D_i = \Phi(z_{1i} + z_{2i} + \epsilon_i). \quad (38)$$

The treatment is continuous and lies in $(0, 1)$.

The structural outcome is generated according to

$$Y_i = 1 + D_i + 5 \sum_{j=1}^7 X_{ji} + D_i u_i. \quad (39)$$

The outcome contains $X_{1i}, \dots, X_{7i}$; together with the controls in the instruments, $X_{1i}, \dots, X_{10i}$ form the ten-control oracle set. For $d \in (0, 1)$, Equation (39) implies

$$Y_i(d) = 1 + 5 \sum_{j=1}^7 X_{ji} + d(1 + u_i). \quad (40)$$

Because u_i is independent of X_i and standard normal,

$$Q_{Y_i(d)|X_i}(\tau) = 1 + 5 \sum_{j=1}^7 X_{ji} + d [1 + \Phi^{-1}(\tau)]. \quad (41)$$

Here Φ^{-1} denotes the standard-normal quantile function, i.e. the inverse of the standard-normal cumulative distribution function Φ . The structural quantile treatment effect is therefore

$$\alpha_0(\tau) = 1 + \Phi^{-1}(\tau). \quad (42)$$

In particular, $\alpha_0(0.50) = 1$.

2. The displayed Monte Carlo equations in Chen et al. (2021) write the corresponding control terms as x_j . The released simulation code instead uses the transformed controls $X_j = \Phi(x_j)$. The simulations in this thesis follow that executable implementation.

The original high-dimensional design of Chen, Huang, and Tien fixes instrument relevance. The thesis instead requires a controlled experiment that weakens the excluded-instrument signal while preserving other economically important features. Simply replacing $z_{1i} + z_{2i}$ by $\kappa(z_{1i} + z_{2i})$ would also reduce latent-index variance. Since $D_i = \Phi(d_i^*)$, this would change the marginal treatment distribution and, through the outcome equation, confound the interpretation of performance differences. An independent Gaussian component compensates for this lost variance.

The extension varies excluded-instrument relevance through $\kappa \in [0, 1]$. Let $w_i \sim \mathcal{N}(0, 1)$ be independent of all preceding primitives. The latent index and treatment are

$$d_i^*(\kappa) = \kappa(z_{1i} + z_{2i}) + \sqrt{2(1 - \kappa^2)} w_i + \epsilon_i, \quad (43)$$

$$D_i(\kappa) = \Phi(d_i^*(\kappa)). \quad (44)$$

This symmetric loading is the sole structural modification to the executable baseline. The reason for the compensating coefficient is that independent standard-normal z_{1i} and z_{2i} give their sum variance two, whereas $\kappa(z_{1i} + z_{2i})$ has variance $2\kappa^2$. An independent standard-normal w_i with coefficient $c(\kappa)$ restores the original variance when $2\kappa^2 + c(\kappa)^2 = 2$. Taking the nonnegative root gives $c(\kappa) = \sqrt{2(1 - \kappa^2)}$. The coefficient therefore follows from variance preservation rather than an arbitrary normalization.

At $\kappa = 1$, the coefficient on w_i is zero and

$$D_i(1) = \Phi(z_{1i} + z_{2i} + \epsilon_i), \quad (45)$$

which reproduces Equation (38) and provides a baseline equivalence check. As κ decreases, the direct contribution of the excluded z components falls and independent w_i supplies more of the same index variance. Conceptually, at $\kappa = 0$ the z components disappear completely while index variance and the endogeneity channel remain unchanged. The experiment stops at $\kappa = 0.10$ to study progressively weak but nonzero designed relevance; this setting does not itself establish formal non-identification. Independence and unit variances imply

$$\text{Var}[d_i^*(\kappa)] = 2\kappa^2 + 2(1 - \kappa^2) + 1 = 3. \quad (46)$$

Thus $d_i^*(\kappa) \sim \mathcal{N}(0, 3)$ for every κ , preserving the marginal distribution of $D_i(\kappa)$. This distribution is not uniform because Φ is applied to an index with variance three.

Instrument relevance changes because

$$\text{Cov}(Z_{1i}, d_i^*(\kappa)) = \text{Cov}(Z_{2i}, d_i^*(\kappa)) = \kappa. \quad (47)$$

Because all other instrument components are independent of $d_i^*(\kappa)$, each instrument–index covariance falls as κ decreases. Only the excluded z component of each observed instrument enters the index directly, so this covariance changes one-for-one with κ . The parameter governs relevance but is not a complete measure of quantile-specific IVQR identification. Because the observed instruments also contain controls and independent noise, κ is neither a conventional first-stage coefficient nor a universal weak-IV statistic. First-stage F and partial R^2 are reported later as descriptive calibration diagnostics; no Stock–Yogo or $F < 10$ classification is imposed on IVQR. Endogeneity remains fixed because u_i is independent of z_{1i} , z_{2i} , and w_i while its covariance with ϵ_i is unchanged:

$$\text{Cov}(u_i, d_i^*(\kappa)) = \text{Cov}(u_i, \epsilon_i) = 0.3. \quad (48)$$

Joint normality and the invariant variance and covariance in Equations (46) and (48) keep the joint distribution of $(u_i, d_i^*(\kappa))$ unchanged. The controls, observed-instrument construction, structural outcome equation, induced marginal outcome distribution, and structural quantile effect are also unchanged. In particular, the 100-dimensional control distribution, construction of Z_{1i} and Z_{2i} , two instrument moments, estimator definitions, and inference procedure remain fixed. Together with $\text{Var}(d_i^*) = 3$ and $\text{Cov}(u_i, d_i^*) = 0.3$, these invariances make the exercise a relevance stress test: the intentionally changing feature is the excluded z components’ contribution to treatment.

This invariance is marginal. Conditioning on the observed instruments changes the conditional treatment-index variance as κ changes, which is inherent to changing the instrument–treatment relationship itself. The design therefore does not claim to hold every conditional dependence measure fixed.

Within each replication, the same primitive draw of u_i , ϵ_i , X_i , z_{1i} , z_{2i} , v_{1i} , v_{2i} , and w_i is reused across all four κ values. This common-random-number construction makes matched comparisons more informative because strength differences are not generated by unrelated samples. It does not eliminate Monte Carlo uncertainty.

3.6 Simulation Design and Estimator Comparison

The completed Monte Carlo experiment uses the design dimensions

$$\begin{aligned} n &\in \{500, 1000\}, & \tau &\in \{0.10, 0.25, 0.50, 0.75, 0.90\}, \\ \kappa &\in \{1, 0.50, 0.25, 0.10\}, & R &= 500. \end{aligned} \quad (49)$$

Here n is the Monte Carlo sample size, τ is the quantile index, κ is the instrument-relevance design parameter, and $R = 500$ is the number of Monte Carlo replications in each (n, τ, κ) design. The Cartesian product contains 40 distinct (n, τ, κ) designs, each with 500 replications.

Oracle-GMM uses the known oracle control set; Full-GMM uses all controls without regularization; and DML-IVQR uses regularized high-dimensional nuisances and an orthogonalization-based score construction, with the implementation-specific residualization intercept described in Section 3.3. The design therefore has 120 design–estimator cells. Every comparison uses the truth $\alpha_0(\tau) = 1 + \Phi^{-1}(\tau)$ from Equation (42).

All estimators use the same sample within a replication. DML-IVQR uses the BC pivotal penalty and same-sample procedure in Section 3.3. Inference is conducted at 95%, and power uses the alternatives defined below.

3.7 Performance Measures and Numerical Inference

For a fixed $(n, \tau, \kappa, \text{estimator})$ cell, let $\widehat{\alpha}_r(\tau)$ denote the point estimate in replication r , and let $\alpha_0(\tau)$ be the true value from Equation (42). Replications are indexed by $r = 1, \dots, R$. Suppressing an estimator superscript, point-estimation performance is summarized by

$$\begin{aligned} \text{Bias}(\tau) &= \frac{1}{R} \sum_{r=1}^R [\alpha_0(\tau) - \widehat{\alpha}_r(\tau)], \\ \text{MAE}(\tau) &= \frac{1}{R} \sum_{r=1}^R |\widehat{\alpha}_r(\tau) - \alpha_0(\tau)|, \\ \text{RMSE}(\tau) &= \left\{ \frac{1}{R} \sum_{r=1}^R [\widehat{\alpha}_r(\tau) - \alpha_0(\tau)]^2 \right\}^{1/2}. \end{aligned} \tag{50}$$

Measures are computed separately by estimator and design cell. Positive Bias means estimates lie below the truth on average; MAE records absolute error, and RMSE gives greater weight to large errors.

When the generic statistic W_N from Chapter 3 is evaluated in the Monte Carlo experiment, N equals the design sample size n ; the additional subscript r identifies the Monte Carlo replication. With $c_{0.95}$ denoting the critical value defined below, define

$$C_r(\tau) = \mathbf{1}\{W_{N,r}(\alpha_0(\tau)) \leq c_{0.95}\}, \quad \widehat{\text{Coverage}}(\tau) = \frac{1}{R} \sum_{r=1}^R C_r(\tau). \tag{51}$$

Coverage evaluates $W_{N,r}$ directly at $\alpha_0(\tau)$ and is therefore independent of the numerical grid. For DML, the pivotal penalty generates fresh Uniform draws on each evaluation, so a direct

truth evaluation and a stored grid-profile evaluation at the same numerical candidate need not use identical pivotal draws. Reported DML coverage is the repeated-sample acceptance frequency of the direct candidate-specific procedure, not literal membership frequency of the particular stored randomized grid realization. Empirical size is

$$\widehat{\text{Size}}(\tau) = \frac{1}{R} \sum_{r=1}^R \mathbf{1}\{W_{N,r}(\alpha_0(\tau)) > c_{0.95}\} = 1 - \widehat{\text{Coverage}}(\tau). \quad (52)$$

The equality follows from the implemented rejection rule.

Power is evaluated at deliberately false treatment-effect values

$$a_{\Delta}(\tau) = \alpha_0(\tau) + \Delta, \quad \Delta \in \{-1.00, -0.50, -0.25, 0.25, 0.50, 1.00\}. \quad (53)$$

For each offset, empirical power is

$$\widehat{\text{Power}}(\Delta, \tau) = \frac{1}{R} \sum_{r=1}^R \mathbf{1}\{W_{N,r}(a_{\Delta}(\tau)) > c_{0.95}\}. \quad (54)$$

The statistic is evaluated directly at each false value, independently of the confidence-region grid; $\Delta = 0$ is size, not power.

The numerical confidence-set calculation proceeds by test inversion (Chernozhukov and Hansen 2008). For fixed τ and estimator, take a candidate structural effect a , re-estimate the required nuisance quantities conditional on it, construct the estimator-specific IVQR score, and compute $W_N(a)$. Under the null that a is the true effect, the asymptotic reference distribution has two degrees of freedom because there are two instrument moment components. Collecting every candidate that passes its own test gives the confidence region. It is not chosen around the point estimate: minimizing $W_N(a)$ and retaining candidates below a critical value are distinct operations.

Conventional Wald inference centered on an estimate typically relies more directly on regular local identification. Direct testing of each hypothesized effect remains useful under weak or partial identification. This robustness concerns test validity, not narrow regions: many competing effects may survive under weak relevance.

The accepted-set measure totals all accepted components inside the primary computational domain, including disconnected components. It is neither the span between extreme accepted values nor an unrestricted confidence-region length. Because $d_z = 2$, the 95% cutoff for the statistic in Section 3.4 is

$$c_{0.95} = \chi_{2,0.95}^2 = 5.991464547 \dots \quad (55)$$

A candidate is accepted when $W_N(a) \leq c_{0.95}$ and rejected when $W_N(a) > c_{0.95}$, with the 95% cutoff 5.991464547... above. This asymptotic cutoff is not an exact finite-sample guarantee. The primary grid is

$$\begin{aligned} A_0 &= [-1, 3], & h &= 0.1, \\ G_0 &= \{a_j = -1 + (j - 1)h : j = 1, \dots, 41\}, \\ &= \{-1, -0.9, \dots, 2.9, 3\}. \end{aligned} \tag{56}$$

Here $A_0 = [-1, 3]$ is the fixed primary computational domain, $h = 0.1$ is the distance between adjacent grid points, j is the grid-point index, and a_j denotes the j -th candidate treatment-effect value. The set G_0 contains all 41 candidate values. There are 41 points because the four-unit interval from -1 to 3 , evaluated in increments of 0.1 and including both endpoints, contains $4/0.1 + 1 = 41$ grid points. The original authors' main Monte Carlo implementation uses `seq(-1, 3, length=41)`. Preserving its four-unit domain and 0.10 resolution avoids choosing a new grid after observing the extension results. All five quantiles, strength settings, and estimators use this common grid, making their measures comparable without differences in grid density. The resolution is inherited, not claimed to be asymptotically optimal. All five true effects lie strictly inside the domain: the extreme values are $\alpha_0(0.10) \approx -0.282$ and $\alpha_0(0.90) \approx 2.282$, with $\alpha_0(0.50) = 1$.

The distinction between the theoretical region and its finite-grid implementation is important. The theoretical region $C = \{a : W_N(a) \leq c_{0.95}\}$ ranges over the relevant parameter space. Computation evaluates only finitely many candidates, producing an accepted set inside A_0 , a computational search region rather than an asserted parameter space. The reported summary is therefore the grid-based accepted-set measure within the primary computational domain $A_0 = [-1, 3]$, rather than unrestricted confidence-region length.

Total confidence-region length is an intuitive informativeness summary, but numerical measurement requires localization inside a sufficiently wide search domain. The initial design considered adaptive expansion; frozen diagnostics showed that weak-relevance accepted sets could keep reaching wider boundaries. Stopping at an arbitrary endpoint and reporting that finite span would make length partly an endpoint artifact. The main comparison therefore uses fixed A_0 , while separate $A_1 = [-3, 5]$ and $A_2 = [-5, 7]$ diagnostics examine endpoint sensitivity.

For replication r , define the binary grid-acceptance indicator, accepted grid, and all-grid-points indicator by

$$\begin{aligned} I_r(a_j) &= \mathbf{1}\{W_{N,r}(a_j) \leq c_{0.95}\}, \\ C_{r,G_0} &= \{a_j \in G_0 : I_r(a_j) = 1\}, \\ F_r &= \mathbf{1}\{I_r(a_j) = 1 \text{ for every } a_j \in G_0\}. \end{aligned} \quad (57)$$

For the primary domain A_0 , define left-boundary, right-boundary, and either-boundary acceptance in replication r by

$$B_r^L = I_r(a_1), \quad B_r^R = I_r(a_{41}), \quad B_r^E = \mathbf{1}\{B_r^L = 1 \text{ or } B_r^R = 1\}. \quad (58)$$

The corresponding either-boundary acceptance rate over R Monte Carlo replications is

$$\widehat{B}^E = \frac{1}{R} \sum_{r=1}^R B_r^E. \quad (59)$$

The same definition is used for an expanded diagnostic domain, with its own left and right endpoints. A boundary is accepted when the candidate-specific test does not reject the endpoint of the finite numerical search interval. A high boundary-acceptance rate therefore means that accepted components frequently reach the edge of the examined domain and are not localized inside it. An accepted endpoint suggests that a component may continue outside the domain; a measure near four indicates limited localization within A_0 . The A_1 and A_2 diagnostics examine whether this behavior disappears. Failure to localize even within A_2 does not establish global unboundedness or failed identification. If boundary acceptance disappears after the domain is widened, the observed accepted grid components no longer touch the investigated endpoints. This does not rule out additional disconnected accepted components outside the investigated domain. Persistent boundary acceptance as successively wider finite domains are examined indicates continued lack of numerical localization over those investigated ranges.

The frozen implementation computes the grid-based accepted-set measure within A_0 as the trapezoidal integral of this binary indicator:

$$M_r(A_0) = \sum_{j=1}^{40} \frac{a_{j+1} - a_j}{2} [I_r(a_j) + I_r(a_{j+1})]. \quad (60)$$

The sum runs from $j = 1$ to 40 because the 41 grid points in G_0 define 40 adjacent grid intervals $[a_j, a_{j+1}]$. The trapezoidal sum includes separated accepted components without filling rejected gaps and does not interpolate critical-value crossings. With $h = 0.10$, an adjacent interval contributes zero if both endpoints are rejected, 0.10 if both are accepted, and 0.05 if

exactly one is accepted. Thus the measure ranges from zero to four and summarizes numerical acceptance, including disconnected components, without solving for exact continuous crossings of $W_N(a)$ with the cutoff. The Monte Carlo mean of F_r is the frequency with which all tested grid points are accepted, not acceptance of the real line.

Coverage evaluates the exact $\alpha_0(\tau)$, whether or not it is a grid point; power evaluates the exact $\alpha_0(\tau) + \Delta$, independently of G_0 . Grid and domain choices affect the accepted-set measure and profile point estimate, but do not mechanically determine reported coverage or power. A post-main diagnostic examines sensitivity to the finite endpoints using

$$A_0 = [-1, 3], \quad A_1 = [-3, 5], \quad A_2 = [-5, 7]. \quad (61)$$

It uses $n = 1000$, 100 saved primitive samples, all five quantiles and estimators, $\kappa \in \{1, 0.25, 0.10\}$, and spacing 0.1. Stage 1 evaluates A_1 and its nested A_0 profile; Stage 2 reloads the same samples and adds only the tails needed for A_2 . Domain comparisons therefore change candidates, not samples. The outputs record accepted measure, grid share and count, boundary acceptance, all-grid acceptance, and incremental measure.

The boundary-acceptance diagnostic follows the finite-domain interpretation of Equations (58)–(59). The bounded diagnostic ends at A_2 , while main summaries retain A_0 and direct coverage and power remain unchanged.

4 Results

The analysis proceeds in four steps: verifying the relevance ordering induced by κ and examining point-estimation accuracy; assessing finite-sample coverage; evaluating accepted-set informativeness and power; and comparing quantiles and estimators. The central question is how quickly deterioration appears, and through which dimensions, in this high-dimensional IVQR design. Reading these measures together distinguishes inaccurate estimation, poorly calibrated inference, and tests that retain too many competing effects to be informative.

4.1 Instrument Relevance and Point Estimation

Table 1 establishes that the four values of κ produce clearly ordered excluded-instrument relevance settings, using the first-stage F -statistic and partial R^2 as descriptive diagnostics.

Both diagnostics fall strongly as κ decreases. For $n = 500$, the median F -statistic falls from 101.17 at $\kappa = 1$ to 18.71 at $\kappa = 0.50$, 4.74 at $\kappa = 0.25$, and 1.22 at $\kappa = 0.10$; the corresponding endpoints for $n = 1000$ are 212.82 and 2.04. Partial R^2 is comparatively stable across sample sizes, whereas F increases with n . The design therefore varies the instruments' explanatory contribution while sample size affects the amount of information available. These diagnostics describe relative relevance within the experiment; they are not quantile-specific weak-IV thresholds.

Table 1. Instrument-strength calibration

n	κ	Median F	F : p10–p90	Median partial R^2	Partial R^2 : p10–p90
500	1.00	101.17	85.93–126.68	0.294	0.261–0.342
500	0.50	18.71	11.99–27.95	0.071	0.047–0.103
500	0.25	4.74	1.74–8.65	0.019	0.007–0.034
500	0.10	1.22	0.24–3.65	0.005	0.001–0.015
1000	1.00	212.82	185.34–239.50	0.301	0.273–0.327
1000	0.50	38.65	30.13–53.60	0.073	0.058–0.098
1000	0.25	9.85	5.34–16.15	0.020	0.011–0.032
1000	0.10	2.04	0.38–5.42	0.004	0.001–0.011

Note: Descriptive Monte Carlo strength diagnostics from the preserved 100-replication calibration. The reported dispersion ranges are the 10th and 90th percentiles. These statistics are not formal weak-instrument classification thresholds.

The two diagnostics separate signal strength from sample information. Partial R^2 falls monotonically from approximately 0.29 and 0.30 at strong relevance to 0.005 and 0.004 at the weakest setting for the smaller and larger samples, respectively. Its relative stability across sample sizes reflects a similar underlying explanatory share, while its decline across κ indicates that the designed population signal is changing. The F -statistic rises with sample size because the same explanatory share is supported by more observations and estimated more precisely. The clean ordering at both sample sizes confirms that the intervention works as designed, without supplying formal IVQR identification thresholds.

Table 2 is the central finite-sample performance table. It reports Bias, MAE, RMSE, Coverage, and the median accepted-set measure for $n = 1000$, $\tau \in \{0.10, 0.50, 0.90\}$, all four κ values, and Oracle-GMM, Full-GMM, and DML-IVQR.

As instrument relevance declines, MAE and RMSE generally increase for all three estimators and across the reported quantiles. At $\tau = 0.50$, moving from $\kappa = 1$ to $\kappa = 0.10$ raises RMSE from 0.072 to 0.945 for Oracle, from 0.100 to 1.018 for Full, and from 0.104 to 0.991 for DML. Oracle deterioration shows that the phenomenon is not generated solely by high-dimensional nuisance estimation: knowing the relevant controls does not replace instrument information. Under the frozen convention $\text{Bias} = \alpha_0 - \widehat{\alpha}$, positive Bias means that estimates are below the truth on average.

The progression is visible before the weakest setting. At $\kappa = 1$, estimates are relatively concentrated, particularly at the median. At $\kappa = 0.50$, deterioration is already apparent, although useful signal remains. By $\kappa = 0.25$, MAE and RMSE have increased markedly; at $\kappa = 0.10$, errors are large across estimators. Modest Bias alongside much larger MAE and RMSE means that positive and negative replication errors partly cancel. Weak relevance therefore manifests importantly as dispersion rather than necessarily a common directional displacement.

Oracle-GMM knows the designated relevant control set and removes the high-dimensional variable-selection problem by construction. Its deterioration provides a benchmark for separating the common relevance problem from additional high-dimensional estimation difficulties. The shared loss of accuracy cannot be attributed solely to Lasso, nuisance selection, or DML regularization: all three procedures face reduced information from the excluded instruments.

4.2 Coverage and Finite-Sample Calibration

Figure 1 and the Coverage column of Table 2 show that empirical coverage is not uniformly close to the nominal level of 0.95. Coverage is evaluated directly at the true structural quantile

Table 2. Main finite-sample performance at $n = 1000$

τ	κ	Estimator	Bias	MAE	RMSE	Coverage	Median accepted-set measure
0.10	1.00	Oracle-GMM	-0.054	0.093	0.121	0.860	0.400
		Full-GMM	-0.329	0.337	0.387	0.502	0.800
		DML-IVQR	-0.005	0.117	0.150	0.968	0.600
	0.50	Oracle-GMM	-0.101	0.194	0.287	0.926	1.000
		Full-GMM	-0.407	0.465	0.630	0.804	1.900
		DML-IVQR	-0.159	0.306	0.482	0.948	2.225
	0.25	Oracle-GMM	-0.384	0.566	0.934	0.930	3.450
		Full-GMM	-0.773	0.914	1.242	0.932	3.500
		DML-IVQR	-0.681	0.844	1.229	0.950	3.750
	0.10	Oracle-GMM	-0.747	0.975	1.376	0.946	4.000
		Full-GMM	-1.074	1.251	1.582	0.942	3.900
		DML-IVQR	-0.932	1.146	1.514	0.936	4.000
0.50	1.00	Oracle-GMM	0.007	0.047	0.072	0.872	0.200
		Full-GMM	0.007	0.072	0.100	0.896	0.400
		DML-IVQR	0.070	0.079	0.104	0.776	0.300
	0.50	Oracle-GMM	0.015	0.093	0.134	0.936	0.500
		Full-GMM	0.014	0.151	0.199	0.946	0.800
		DML-IVQR	0.059	0.119	0.160	0.918	0.600
	0.25	Oracle-GMM	0.045	0.281	0.431	0.946	1.900
		Full-GMM	0.029	0.372	0.500	0.936	2.200
		DML-IVQR	-0.003	0.306	0.448	0.948	2.025
	0.10	Oracle-GMM	-0.044	0.735	0.945	0.942	3.900
		Full-GMM	-0.012	0.840	1.018	0.948	3.800
		DML-IVQR	-0.195	0.779	0.991	0.942	3.900
0.90	1.00	Oracle-GMM	0.072	0.110	0.148	0.904	0.500
		Full-GMM	0.387	0.397	0.465	0.530	1.000
		DML-IVQR	0.183	0.186	0.215	0.498	0.400
	0.50	Oracle-GMM	0.211	0.295	0.540	0.940	1.575
		Full-GMM	0.584	0.629	0.871	0.834	2.450
		DML-IVQR	0.226	0.260	0.396	0.844	1.100
	0.25	Oracle-GMM	0.491	0.658	1.038	0.952	3.650
		Full-GMM	0.901	1.058	1.389	0.920	3.600
		DML-IVQR	0.417	0.580	0.927	0.922	3.350
	0.10	Oracle-GMM	0.653	0.948	1.350	0.964	4.000
		Full-GMM	1.050	1.236	1.583	0.948	3.900
		DML-IVQR	0.519	0.831	1.191	0.940	4.000

Note: Each cell is based on 500 Monte Carlo replications. Bias follows the frozen convention ($\text{Bias} = \alpha_0 - \widehat{\alpha}$). Coverage is evaluated directly at the true structural quantile effect. The accepted-set measure is the grid-based measure within the prespecified primary computational domain ($A_0 = [-1, 3]$); it is not an unrestricted confidence-region length.

effect, independently of the numerical grid. The preserved procedure uses an asymptotic chi-square reference cutoff; its limiting validity for this DGP is not established here.

Because each reported coverage rate is based on $R = 500$ Monte Carlo replications, the simulation itself introduces sampling uncertainty. For an estimated rejection or coverage probability $\hat{p}$, the Monte Carlo standard error is

$$\text{MCSE}(\hat{p}) = \sqrt{\frac{\hat{p}(1 - \hat{p})}{R}}.$$

With $R = 500$, the largest possible MCSE is approximately 0.022, attained near $\hat{p} = 0.5$; at the nominal coverage probability 0.95, the corresponding MCSE is approximately 0.010. Consequently, very small differences between nearby coverage rates should not be given substantive weight. The large deviations emphasized here are materially larger than Monte Carlo uncertainty: coverage values around 0.50, 0.63, or 0.86, for example, remain far from the nominal 0.95 benchmark relative to their simulation standard errors.

Differences of one or two percentage points warrant caution, whereas deviations of tens of percentage points from nominal coverage are far too large to be explained by the sampling uncertainty associated with $R = 500$ replications.

Distortions occur even at $\kappa = 1$ and vary substantially by estimator and quantile. For $n = 1000$, Full coverage is 0.502 at $\tau = 0.10$ and 0.530 at $\tau = 0.90$. DML coverage is 0.632 at $\tau = 0.75$ and 0.498 at $\tau = 0.90$, but 0.968 at $\tau = 0.10$. Oracle coverage is 0.860 at both $\tau = 0.10$ and $\tau = 0.75$, showing that high-dimensional nuisance estimation is not the sole explanation for undercoverage.

Coverage should be close to 0.95 when the asymptotic chi-square distribution approximates the finite-sample null distribution well. The strong-relevance differences instead reveal calibration heterogeneity across estimator–quantile combinations. Increasing the sample size does not uniformly remove these distortions: the larger sample still contains substantial undercoverage. Small differences between the two sample sizes are less informative than the persistence of large deviations.

Calibration and weak identification are distinct issues. The robust test construction avoids requiring a strongly identified point estimator in the usual way, but still uses an asymptotic chi-square cutoff. The implemented tests using the asymptotic chi-square reference cutoff are imperfectly calibrated in several simulated cells. The experiment does not distinguish finite-sample approximation error from departures of the preserved executable procedure or this DGP from the regularity conditions underlying the idealized asymptotic theory. Undercover-

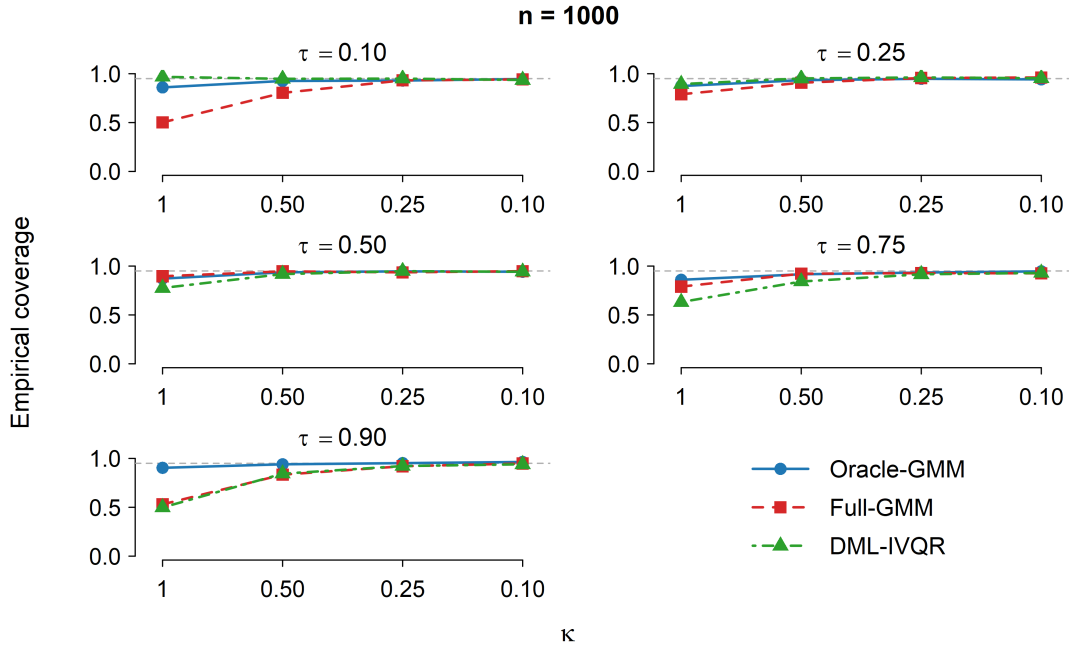
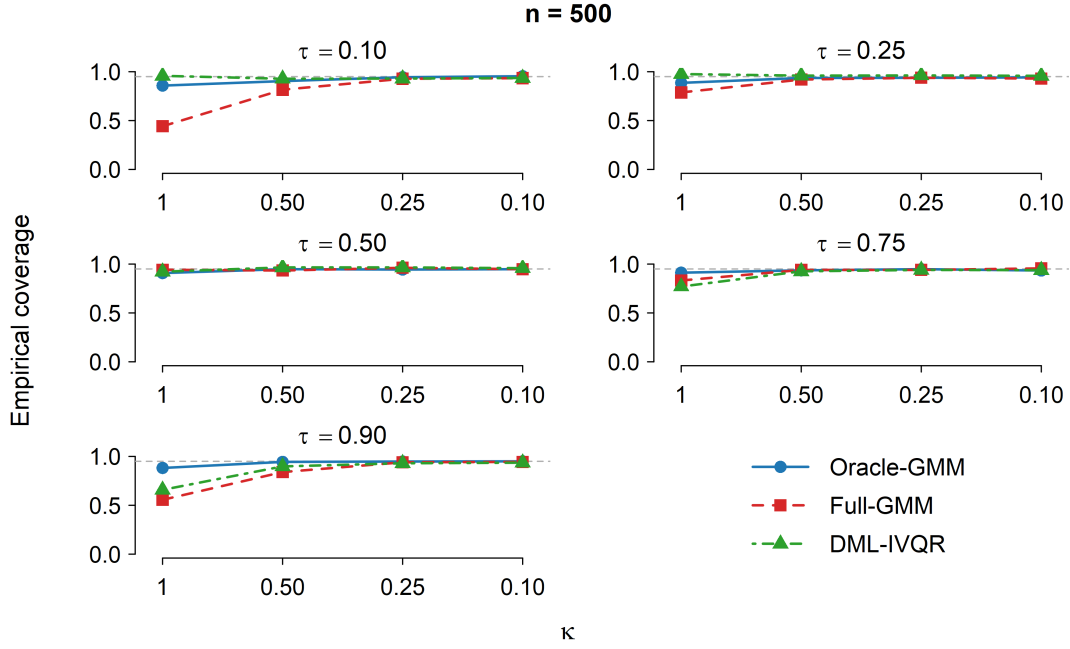


Figure 1. Empirical coverage across instrument-strength designs.

age already present with strong relevance therefore cannot be attributed to weak identification alone.

As relevance becomes very weak, the test may reject few candidates, retaining the truth frequently while accepted sets cover most of the primary domain. Such high coverage measures calibration and can coexist with very low informativeness and power. An increase in coverage as relevance weakens is therefore not automatically an improvement: truth retention may rise because the procedure rejects fewer values generally. Coverage must be read alongside the evidence on exclusion of false candidates.

4.3 Confidence-Set Informativeness and Power

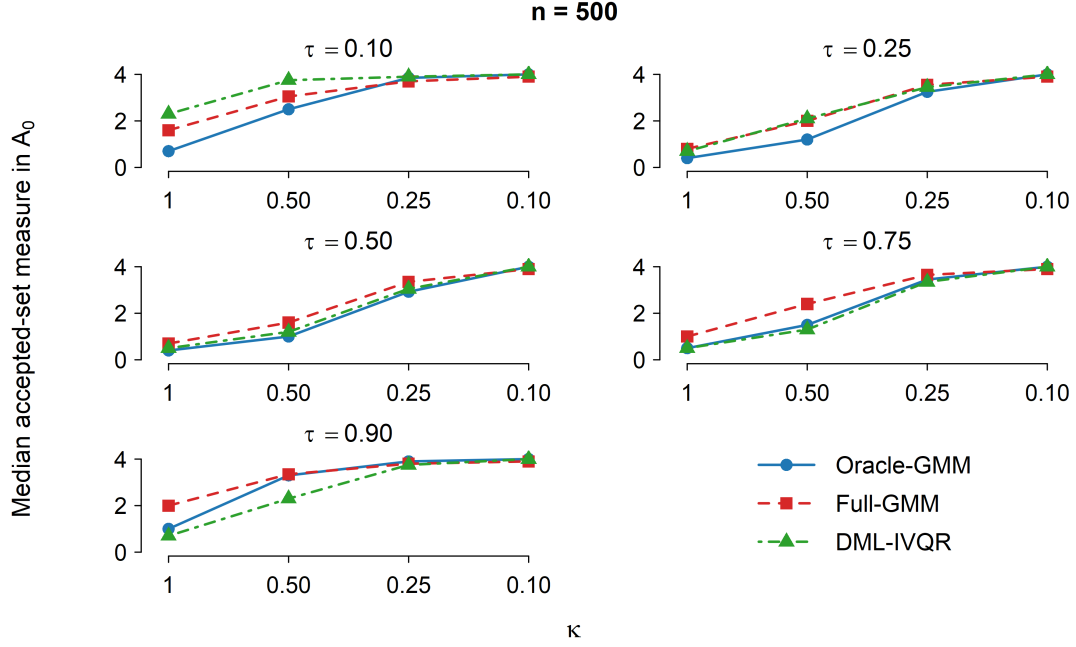
Figure 2 shows the median grid-based accepted-set measure within $A_0 = [-1, 3]$. Panels (a) and (b) report $n = 500$ and $n = 1000$, respectively, allowing comparison across sample sizes for the same quantiles and estimators. The plotted quantity measures acceptance inside the four-unit primary domain, not unrestricted confidence-region length.

The measure rises strongly as κ falls, and at $\kappa = 0.10$ accepted sets often occupy nearly the whole primary domain. For $n = 1000$ and $\tau = 0.50$, the Oracle, Full, and DML medians rise respectively from 0.200, 0.400, and 0.300 at $\kappa = 1$ to 3.900, 3.800, and 3.900 at $\kappa = 0.10$. Increasing n generally improves localization, but does not solve the weakest-relevance case. The accepted-set measure is deliberately defined on the fixed primary domain $A_0 = [-1, 3]$. The separate domain-expansion diagnostic and the choice of the primary domain are discussed in Appendix A.1.

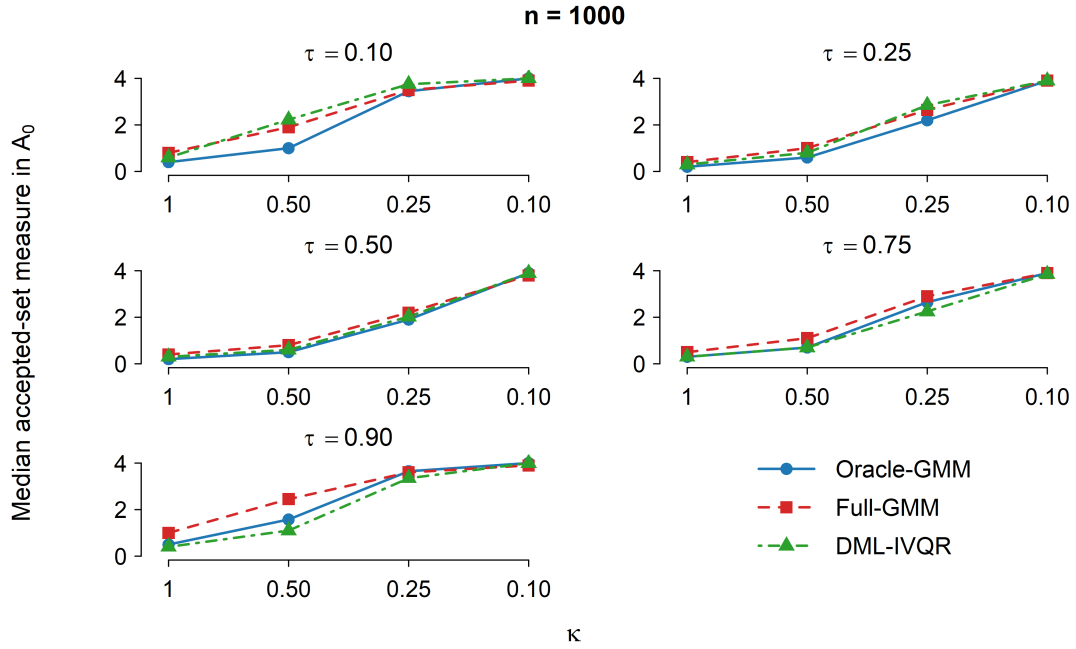
Small accepted-set measures under strong relevance indicate that many competing candidates are excluded. Expansion at intermediate relevance reflects a progressively less localized effect. A value near four means that almost the entire primary domain survives testing, not that the unrestricted confidence region has been shown to have length four. It is direct evidence of low localization within A_0 .

The sample-size comparison is most favorable where useful instrument signal remains: additional observations generally narrow the accepted set. At the weakest relevance, however, measures remain large even in the larger sample. More observations help, but do not fully compensate for the information deficit at these simulated sizes; two sample sizes do not establish an asymptotic conclusion.

Figure 3 reports rejection probabilities under moderate false alternatives for $n = 1000$: panel (a) uses $\Delta = -0.50$ and panel (b) uses $\Delta = +0.50$. Rejection probabilities fall strongly as κ decreases, so false candidate treatment effects survive more frequently.



(a) $n = 500$



(b) $n = 1000$

Figure 2. Grid-based accepted-set measure within the primary computational domain $A_0 = [-1, 3]$.

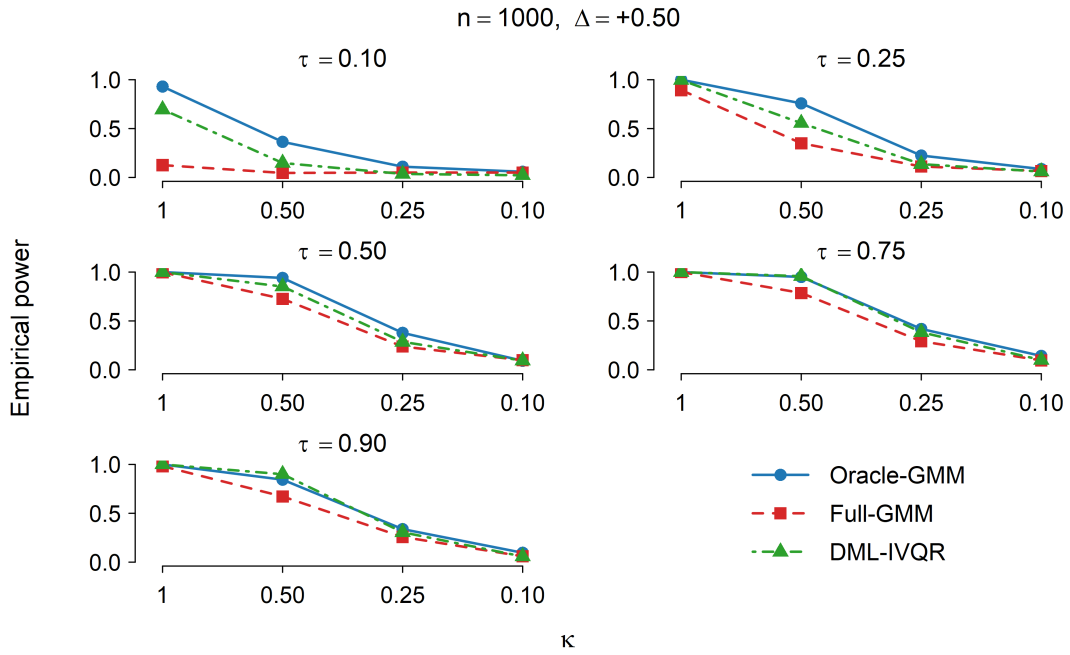
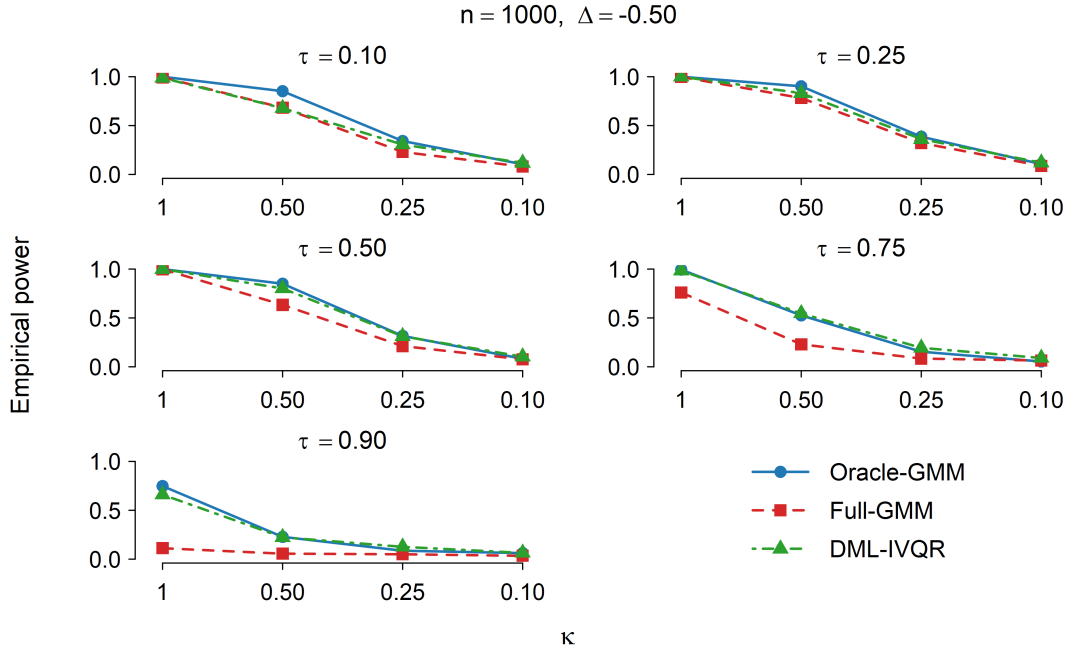


Figure 3. Rejection probability under moderate false alternatives, $n = 1000$.

The median-quantile results show a substantial loss of discrimination even at intermediate relevance. At $\tau = 0.50$ and $\Delta = -0.50$, power falls from 1.000 through 0.850 and 0.316 to 0.082 for Oracle, and from 1.000 through 0.802 and 0.312 to 0.104 for DML across the four declining relevance settings. Full power falls from 0.998 to 0.076 between the strongest and weakest settings. At the positive alternative, the corresponding weakest-relevance probabilities are 0.092, 0.098, and 0.094. Power is the fraction of replications rejecting the specified false effect $\alpha_0(\tau) + \Delta$; its decline directly records a reduced ability to distinguish that effect from the truth. Positive and negative displacements need not have equal power because IVQR moments are nonlinear in the candidate parameter. The finite-sample criterion need not be symmetric around the truth, so equal displacements can produce different rejection probabilities. This asymmetry alone is not evidence of point-estimation bias. These are rejection probabilities at a common nominal chi-square cutoff, not size-adjusted power comparisons. Because empirical null rejection rates differ across estimators and designs, cross-estimator differences in raw power should not be interpreted as pure efficiency rankings.

Figures 2 and 3 reveal the same loss of information from complementary perspectives: accepted sets ask which candidates survive, while power asks whether specific false candidates are rejected. As relevance falls, survival expands and rejection declines. Their agreement is the clearest evidence here that weak relevance reduces inferential informativeness even with a test construction designed for weak-identification robustness.

4.4 Quantile Heterogeneity and Estimator Comparison

Taken together, Table 2 and Figures 1–3 show that weakening relevance worsens performance across quantiles, although the magnitude differs. Tail quantiles often display larger estimation errors or more severe calibration problems, but there is no universal ordering between the center and tails or between the two tails.

The same relevance setting yields different accuracy, coverage, and power across quantiles. Quantile heterogeneity may reflect differences in local score behavior, nuisance estimation, density features, and finite-sample information, but the experiment does not identify a single causal mechanism. Tail deterioration is common, but the five quantiles have no universal monotone performance ranking.

Oracle generally provides the strongest benchmark, yet deteriorates sharply as relevance weakens. Full estimates all high-dimensional controls without regularization and often has larger error or lower power. DML often improves on Full, but does not dominate it in every cell or reproduce Oracle uniformly. Its substantial coverage distortions in some quantiles also

show that gains in accuracy need not imply gains in calibration. These differences cannot be attributed solely to regularization because the estimators also use different residualization conventions.

The experiment answers the research question along four dimensions. Weaker relevance substantially worsens point-estimation accuracy; accepted candidates expand and power falls, revealing reduced informativeness. These effects also occur for Oracle, so they are not solely high-dimensional nuisance-estimation phenomena. Coverage adds a separate calibration dimension that varies across estimator and quantile and does not move mechanically with informativeness. DML can reduce some high-dimensional nuisance-estimation costs, but cannot restore information that weak instruments do not supply.

5 Conclusion

This thesis studies how systematically weakening excluded-instrument relevance affects the finite-sample performance and informativeness of weak-identification-robust inference in high-dimensional IVQR. The Monte Carlo extension varies a single relevance parameter, κ , while preserving the marginal treatment distribution, the endogeneity channel, the structural outcome equation, the true quantile treatment effects, the controls, and the observed instrument construction of the baseline design in Chen et al. (2021). Oracle-GMM, Full-GMM, and DML-IVQR are evaluated on identical simulated samples across five quantiles and two sample sizes. This design isolates the consequences of progressively reducing the contribution of the excluded instruments to treatment without interpreting κ as a complete measure of quantile-specific IVQR identification.

The main result is clear. As excluded-instrument relevance declines, point estimation becomes substantially less accurate and candidate-specific inference becomes less informative. MAE and RMSE increase for all three estimators, including Oracle-GMM. At the median with $n = 1000$, for example, Oracle RMSE rises from 0.072 at $\kappa = 1$ to 0.945 at $\kappa = 0.10$, while DML-IVQR rises from 0.104 to 0.991. The deterioration of the oracle benchmark shows that this pattern is not generated solely by high-dimensional nuisance estimation. Reduced instrument information directly weakens the ability of the IVQR moments to distinguish the true treatment effect from competing candidate values.

The inference results reinforce this conclusion. The grid-based accepted-set measure within the prespecified domain $A_0 = [-1, 3]$ expands sharply as relevance falls, while rejection probabilities against false treatment-effect values decline. At the weakest relevance setting, accepted values frequently occupy almost the entire primary domain and power against moderate false alternatives is low even at $n = 1000$. Larger samples improve localization and rejection power when usable instrument signal remains, but they do not eliminate the loss of informativeness at $\kappa = 0.10$. The magnitude of this deterioration also varies across quantiles: tail quantiles are often more difficult than the median, but there is no universal ordering across all estimators and relevance levels.

A separate finding concerns finite-sample calibration. Empirical coverage is not uniformly close to the nominal 0.95 level, including in some cells with the strongest relevance setting. Undercoverage also appears for Oracle-GMM, which rules out high-dimensional regularization as the sole explanation. With $R = 500$ replications, the maximum Monte Carlo standard error of a reported coverage probability is approximately 0.022, so the large coverage deviations documented in the simulations cannot be explained by Monte Carlo noise alone. The

implemented tests using the asymptotic chi-square reference cutoff are imperfectly calibrated in several simulated cells. The experiment does not distinguish finite-sample approximation error from departures of the preserved executable procedure or this DGP from the regularity conditions underlying the idealized asymptotic theory. At the same time, very high coverage under weak relevance is not necessarily desirable: it can coexist with broad accepted sets and very low power against false candidates.

The estimator comparison clarifies both the usefulness and the limits of DML-IVQR. Regularized nuisance estimation often improves performance relative to unregularized Full-GMM, but DML-IVQR neither dominates in every design cell nor uniformly reproduces the Oracle benchmark. Because the estimators also differ in their residualization implementation, the observed gaps cannot be interpreted as the isolated causal effect of regularization. More fundamentally, Neyman orthogonality protects the target moment against first-order nuisance-estimation error; it does not supply the identifying information lost when excluded instruments become weakly relevant.

The scope of these conclusions is deliberately limited. The analysis uses one preserved DGP, two sample sizes, one same-sample DML-IVQR implementation, a finite numerical candidate domain, and the authors' asymptotic chi-square inference procedure. It does not study alternative instrument structures, cross-fitted implementations, alternative critical-value procedures, or an empirical application. These are natural directions for further work. Applicability of the idealized asymptotic regularity argument to the preserved procedure and this DGP remains unestablished.

The central implication is both practical and methodological. In high-dimensional IVQR, successful nuisance regularization and weak-identification-robust test construction do not by themselves guarantee precise or informative inference. When excluded instruments contain little information about treatment, estimation error can rise sharply, false structural effects may be difficult to reject, and finite-sample calibration can remain imperfect. Applied work should therefore evaluate instrument relevance, point-estimation accuracy, coverage, and inferential informativeness separately. No single measure provides sufficient evidence of reliable structural inference.

References

- Anderson, T. W., and Herman Rubin. 1949. "Estimation of the Parameters of a Single Equation in a Complete System of Stochastic Equations." *The Annals of Mathematical Statistics* 20 (1): 46–63. <https://doi.org/10.1214/aoms/1177730090>. [5]
- Andrews, Isaiah, James H. Stock, and Liyang Sun. 2019. "Weak Instruments in Instrumental Variables Regression: Theory and Practice." *Annual Review of Economics* 11:727–753. <https://doi.org/10.1146/annurev-economics-080218-025643>. [2, 5]
- Belloni, Alexandre, and Victor Chernozhukov. 2011. " ℓ_1 -Penalized Quantile Regression in High-Dimensional Sparse Models." *The Annals of Statistics* 39 (1): 82–130. <https://doi.org/10.1214/10-AOS827>. [5, 13]
- Belloni, Alexandre, Victor Chernozhukov, and Kengo Kato. 2019. "Valid Post-Selection Inference in High-Dimensional Approximately Sparse Quantile Regression Models." *Journal of the American Statistical Association* 114 (526): 749–758. <https://doi.org/10.1080/01621459.2018.1442339>. [5]
- Chen, Jau-er, Chien-Hsun Huang, and Jia-Jyun Tien. 2021. "Debiased/Double Machine Learning for Instrumental Variable Quantile Regressions." *Econometrics* 9 (2): 15. <https://doi.org/10.3390/econometrics9020015>. [1, 2, 5–7, 9, 11–13, 18, 19, 37]
- Chernozhukov, Victor, Denis Chetverikov, Mert Demirer, et al. 2018. "Double/Debiased Machine Learning for Treatment and Structural Parameters." *The Econometrics Journal* 21 (1): C1–C68. <https://doi.org/10.1111/ectj.12097>. [1, 5, 12, 15]
- Chernozhukov, Victor, and Christian Hansen. 2005. "An IV Model of Quantile Treatment Effects." *Econometrica* 73 (1): 245–261. <https://doi.org/10.1111/j.1468-0262.2005.00570.x>. [1, 4, 7]
- Chernozhukov, Victor, and Christian Hansen. 2006. "Instrumental Quantile Regression Inference for Structural and Treatment Effect Models." *Journal of Econometrics* 132 (2): 491–525. <https://doi.org/10.1016/j.jeconom.2005.02.009>. [4]
- Chernozhukov, Victor, and Christian Hansen. 2008. "Instrumental Variable Quantile Regression: A Robust Inference Approach." *Journal of Econometrics* 142 (1): 379–398. <https://doi.org/10.1016/j.jeconom.2007.06.005>. [4, 17, 23]
- Chernozhukov, Victor, Christian Hansen, and Michael Jansson. 2009. "Finite Sample Inference for Quantile Regression Models." *Journal of Econometrics* 152 (2): 93–103. <https://doi.org/10.1016/j.jeconom.2009.01.004>. [5, 18]
- Chernozhukov, Victor, Christian Hansen, and Kaspar Wüthrich. 2017. "Instrumental Variable Quantile Regression." In *Handbook of Quantile Regression*, edited by Roger Koenker, Victor Chernozhukov, Xuming He, and Limin Peng, 119–143. Boca Raton, FL: CRC Press. <https://doi.org/10.1201/9781315120256>. [4]
- Jin, Zequn, and Jisheng Sun. 2025. "Neyman-Orthogonal Moment for Instrumental Variable Quantile Regression Model with High Dimensional Data." *Economics Letters* 253:112369. <https://doi.org/10.1016/j.econlet.2025.112369>. [6]
- Kleibergen, Frank. 2002. "Pivotal Statistics for Testing Structural Parameters in Instrumental Variables Regression." *Econometrica* 70 (5): 1781–1803. <https://doi.org/10.1111/1468-0262.00353>. [5]
- Koenker, Roger, and Gilbert Bassett Jr. 1978. "Regression Quantiles." *Econometrica* 46 (1): 33–50. <https://doi.org/10.2307/1913643>. [4]
- Moreira, Marcelo J. 2003. "A Conditional Likelihood Ratio Test for Structural Models." *Econometrica* 71 (4): 1027–1048. <https://doi.org/10.1111/1468-0262.00438>. [5]

- Staiger, Douglas, and James H. Stock. 1997. "Instrumental Variables Regression with Weak Instruments." *Econometrica* 65 (3): 557–586. <https://doi.org/10.2307/2171753>. [2, 5]
- Stock, James H., and Jonathan H. Wright. 2000. "GMM with Weak Identification." *Econometrica* 68 (5): 1055–1096. <https://doi.org/10.1111/1468-0262.00151>. [2, 5]
- Wüthrich, Kaspar. 2019. "A Closed-Form Estimator for Quantile Treatment Effects with Endogeneity." *Journal of Econometrics* 210 (2): 219–235. <https://doi.org/10.1016/j.jeconom.2018.11.017>. [6]

Appendix A Numerical-Domain Diagnostics

A.1 Choice of the Primary Computational Domain

The primary computational domain is $A_0 = [-1, 3]$, evaluated on 41 equally spaced candidate values with step size $h = 0.1$. This domain is retained from the preserved Monte Carlo implementation rather than chosen after observing the results of the weak-instrument extension. Using the same domain for all sample sizes, quantiles, instrument-relevance settings, and estimators makes the accepted-set measures directly comparable across the simulation design.

The interval also contains the true structural quantile effects for all five quantiles considered in the thesis. Under the data-generating process,

$$\alpha_0(\tau) = 1 + \Phi^{-1}(\tau),$$

so the true effects are approximately $-0.282, 0.326, 1.000, 1.674$, and 2.282 for $\tau = 0.10, 0.25, 0.50, 0.75$, and 0.90 , respectively. Thus, none of the true parameter values lies close to or outside the boundaries of A_0 . The interval should therefore be understood as a common numerical search domain, not as an assumed theoretical parameter space.

A finite search domain can nevertheless become restrictive when instruments are weak and many candidate values are accepted. For this reason, a separate sensitivity diagnostic expands the search range from $A_0 = [-1, 3]$ to $A_1 = [-3, 5]$ and then to $A_2 = [-5, 7]$. Under strong instrument relevance, expanding the domain adds little accepted measure, indicating that the observed accepted components are well localized within the investigated ranges. At intermediate relevance, the wider domains contain additional accepted values in some profiles. Under the weakest relevance setting, the accepted measure continues to increase as the domain is expanded and does not stabilize within A_2 .

These results clarify the interpretation of the main accepted-set measure. A value close to four in A_0 means that almost the entire four-unit primary domain is accepted; it does not mean that the unrestricted confidence region has length four. Similarly, continued acceptance in the wider domains indicates poor numerical localization over the ranges examined, but it does not by itself establish that the theoretical confidence region is globally unbounded or that identification completely fails.

Table A.1 retains the key frozen domain-measure comparisons; Figure A.1 summarizes the expansion pattern.

Table A.1. Grid-domain diagnostic: accepted-set measures

n	τ	κ	Estimator	Med. A_0	Med. A_1	Med. A_2	Mean A_0	Mean A_1	Mean A_2
1000	0.10	1.00	Oracle-GMM	0.400	0.400	0.400	0.386	0.386	0.386
1000	0.10	1.00	Full-GMM	0.800	0.800	0.800	0.757	0.757	0.757
1000	0.10	1.00	DML-IVQR	0.600	0.600	0.600	0.705	0.705	0.705
1000	0.10	0.25	Oracle-GMM	3.400	5.325	7.250	3.019	4.994	6.838
1000	0.10	0.25	Full-GMM	3.375	5.950	8.200	3.304	5.774	8.037
1000	0.10	0.25	DML-IVQR	3.750	5.950	7.950	3.607	6.040	8.245

Continued on next page

TableA.1. Grid-domain diagnostic: accepted-set measures (Continued)

n	τ	κ	Estimator	Med. A_0	Med. A_1	Med. A_2	Mean A_0	Mean A_1	Mean A_2
1000	0.10	0.10	Oracle-GMM	4.000	7.900	11.900	3.745	7.381	11.002
1000	0.10	0.10	Full-GMM	3.900	7.800	11.600	3.785	7.394	10.975
1000	0.10	0.10	DML-IVQR	4.000	7.575	10.775	3.845	7.219	10.339
1000	0.25	1.00	Oracle-GMM	0.300	0.300	0.300	0.259	0.259	0.259
1000	0.25	1.00	Full-GMM	0.400	0.400	0.400	0.429	0.429	0.429
1000	0.25	1.00	DML-IVQR	0.300	0.300	0.300	0.346	0.346	0.346
1000	0.25	0.25	Oracle-GMM	2.175	2.800	2.850	2.224	3.158	3.964
1000	0.25	0.25	Full-GMM	2.450	3.500	4.075	2.517	3.768	4.745
1000	0.25	0.25	DML-IVQR	2.750	4.650	6.225	2.630	4.130	5.712
1000	0.25	0.10	Oracle-GMM	3.900	7.700	11.500	3.492	6.699	9.872
1000	0.25	0.10	Full-GMM	3.825	7.600	11.100	3.595	6.891	10.083
1000	0.25	0.10	DML-IVQR	4.000	7.700	11.125	3.651	6.926	9.984
1000	0.50	1.00	Oracle-GMM	0.200	0.200	0.200	0.235	0.235	0.235
1000	0.50	1.00	Full-GMM	0.400	0.400	0.400	0.373	0.373	0.373
1000	0.50	1.00	DML-IVQR	0.300	0.300	0.300	0.259	0.259	0.259
1000	0.50	0.25	Oracle-GMM	1.875	2.200	2.200	1.960	2.682	3.229
1000	0.50	0.25	Full-GMM	2.250	2.400	2.550	2.368	3.171	3.802
1000	0.50	0.25	DML-IVQR	1.950	2.500	3.025	2.124	3.020	3.942
1000	0.50	0.10	Oracle-GMM	3.825	7.600	11.300	3.521	6.640	9.687
1000	0.50	0.10	Full-GMM	3.900	7.625	11.450	3.536	6.718	9.817
1000	0.50	0.10	DML-IVQR	3.900	7.500	10.325	3.538	6.671	9.585
1000	0.75	1.00	Oracle-GMM	0.300	0.300	0.300	0.290	0.290	0.290
1000	0.75	1.00	Full-GMM	0.500	0.500	0.500	0.512	0.512	0.512
1000	0.75	1.00	DML-IVQR	0.300	0.300	0.300	0.279	0.279	0.279
1000	0.75	0.25	Oracle-GMM	2.825	4.200	4.850	2.628	3.945	4.990
1000	0.75	0.25	Full-GMM	2.950	4.450	5.150	2.827	4.267	5.478
1000	0.75	0.25	DML-IVQR	2.325	3.300	3.900	2.402	3.595	4.788
1000	0.75	0.10	Oracle-GMM	3.975	7.750	11.700	3.670	6.999	10.357
1000	0.75	0.10	Full-GMM	3.900	7.750	11.500	3.661	7.023	10.267
1000	0.75	0.10	DML-IVQR	3.900	7.350	10.375	3.554	6.691	9.736
1000	0.90	1.00	Oracle-GMM	0.500	0.500	0.500	0.519	0.519	0.519
1000	0.90	1.00	Full-GMM	1.000	1.000	1.000	1.026	1.031	1.031
1000	0.90	1.00	DML-IVQR	0.400	0.400	0.400	0.454	0.454	0.454
1000	0.90	0.25	Oracle-GMM	3.650	5.850	7.850	3.381	5.625	7.675
1000	0.90	0.25	Full-GMM	3.675	6.150	8.575	3.555	6.123	8.432
1000	0.90	0.25	DML-IVQR	3.450	5.400	7.300	3.114	5.249	7.374
1000	0.90	0.10	Oracle-GMM	4.000	7.900	11.800	3.712	7.241	10.771
1000	0.90	0.10	Full-GMM	3.900	7.700	11.500	3.817	7.448	11.031
1000	0.90	0.10	DML-IVQR	4.000	7.700	11.600	3.644	7.042	10.395

Note: Domains are $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$. Diagnostics concern only these investigated numerical domains and do not establish global boundedness, unboundedness, or disconnected components.

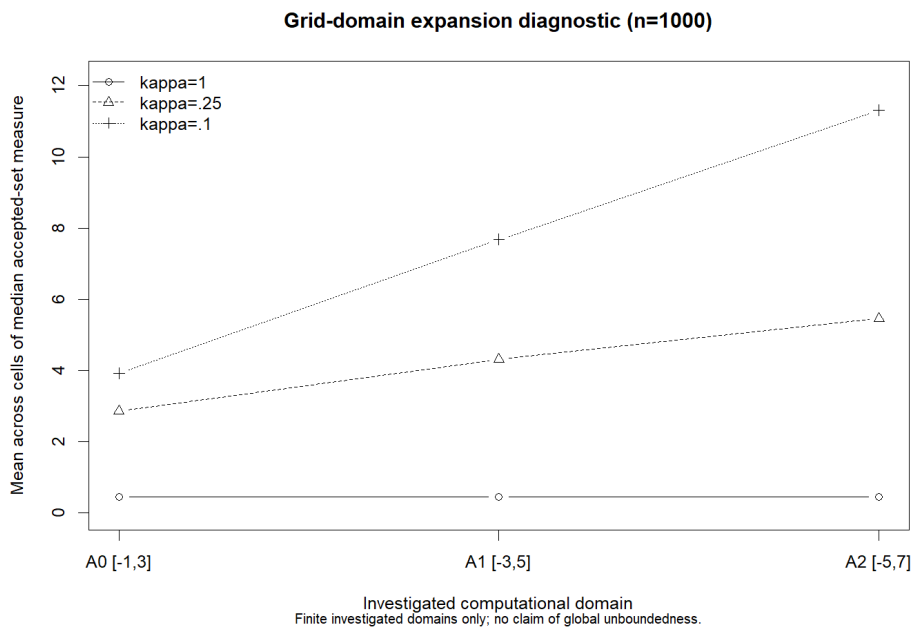


Figure A.1. Sensitivity of the accepted set to expansion of the computational domain

Note: Each plotted value is the mean across 15 estimator–quantile cells of the cell-specific median accepted-set measure for the stated κ and domain. The investigated domains are $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$. The diagnostic records whether the accepted set stabilizes within these numerical domains; it does not establish global boundedness or unboundedness.

A.2 Numerical Handling of the Domain Diagnostic

The diagnostic recorded one Stage 1 numerical failure: replication 35 at $\tau = 0.90$, $\kappa = 0.10$, for DML-IVQR at candidate $a = -2.1$, with a singular-design error. Stage 2 recorded no new evaluation errors. The affected profile was treated as incomplete, and its aggregate accepted-set measures were not calculated from partial information.

Appendix B Coverage Forensic Audit

Coverage is evaluated directly at the exact true structural effect $\alpha_0(\tau)$, independently of whether that value lies on the finite candidate grid. The forensic audit checked 60,000 direct truth evaluations: 500 replications for each of 120 sample-size, quantile, instrument-strength, and estimator cells.

Wilson 95% Monte Carlo intervals exclude nominal 0.95 coverage in 54 of the 120 cells. The audit finds no missing or nonfinite true-parameter W statistics. The large coverage distortions reported in the main text are therefore neither plotting-grid artifacts nor consequences of missing-value handling. The frozen audit report and supporting diagnostics remain preserved in the repository.

Appendix C Reproducibility Note

The preserved author-replication and main-extension environment is R 3.4.3 on 64-bit Windows. The main Monte Carlo uses $R = 500$ replications per design and seed 675. The grid-domain diagnostic also uses seed 675 and reuses saved primitives in its second stage. The separate instrument-strength calibration uses seed 20260813. The frozen reporting outputs are exactly reproducible from the archived CSVs. The parent-process seed records the simulation setup, but because the preserved parallel DML implementation does not explicitly initialize worker-specific RNG streams, exact regeneration of all pivotal penalty draws from the seed alone is not established.

The reporting layer reads frozen simulation, calibration, and diagnostic outputs without re-estimating models. Reporting consistency checks and source/output hash manifests connect those records to the numerical tables. Figure-data checks verify all 480 plotted values against the frozen reporting CSVs, while the figure manifest records source and final-image hashes. Full supplementary tables, RMSE figures, detailed diagnostics, configuration files, and environment records remain in the repository even where they are not printed in the current PDF.

Declaration

I hereby affirm that I have written the above thesis independently and have used no sources or aids other than those indicated; that the submitted work has not previously been submitted for examination at any other university; and that it has not been published, either in whole or in part. I have clearly indicated all direct quotations and passages paraphrased from other works as such.

September 17, 2026

Olimjon Umurzokov